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Data Integration, Sensitivity Analysis and Inverse Problem Formulation for a Compartmental Colon Model

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Abstract

The colonic microbiota ferments the fibre we cannot digest, but the same diet has different effects from one person to another. The only measurement available is the faecal sample, which tells what leaves the colon and not what happens inside it. A mechanistic model can bridge this gap, provided it is connected to real data and calibrated on the individual subject.

In this thesis the carbohydrate degradation model of Muñoz-Tamayo et al., which describes the colon as three regions in series divided into lumen and mucus, is used as a quantitative tool on 3613 faecal metagenomic samples, each summarised by the compositional weights of four functional profiles. The work proceeds in three steps. The four microbial groups of the model, originally defined by the substrate they consume, become populations that spread their activity over several reactions, so that they can be matched with the four profiles. A global sensitivity analysis then identifies the parameters that control the distal microbial composition: a Morris screening reduces thirty factors to twelve, and Sobol' indices show that the block of catabolic fractions absorbs almost all the variance. These are finally estimated sample by sample by solving a Bayesian inverse problem: the likelihood is built on the Aitchison distance, the posterior is sampled with MCMC (DEMetropolisZ), and the forward model is replaced by a neural surrogate.

The observed compositions are reconstructed across the whole dataset, and the estimated parameters remain valid when moving back from the surrogate to the full model. The patient is no longer a datum but a model that can be simulated and perturbed: a first step towards a digital twin of the host-microbiota system.

Keywords: gut microbiota, global sensitivity analysis, Sobol' indices, Bayesian inverse problem, MCMC, neural surrogate model

Abstract in lingua italiana

Il microbiota del colon fermenta le fibre che non riusciamo a digerire, ma la stessa dieta ha effetti diversi da persona a persona. L'unica misura disponibile è il campione fecale, che dice cosa esce dal colon e non cosa accade al suo interno. Un modello meccanicistico può colmare questa distanza, a patto di collegarlo a dati reali e di calibrarlo sul singolo soggetto.

In questa tesi il modello di degradazione dei carboidrati di Muñoz-Tamayo et al., che descrive il colon come tre regioni in serie divise in lume e muco, viene usato come strumento quantitativo su 3613 campioni metagenomici fecali, ciascuno riassunto dai pesi composizionali di quattro profili funzionali. Il lavoro procede in tre passi. I quattro gruppi microbici del modello, definiti in origine dal substrato che consumano, diventano popolazioni che ripartiscono la propria attività su più reazioni, così da corrispondere ai quattro profili. Un'analisi di sensitività globale individua poi i parametri che controllano la composizione microbica distale: uno screening di Morris riduce trenta fattori a dodici, e gli indici di Sobol' mostrano che il blocco delle frazioni cataboliche assorbe quasi tutta la varianza. Queste vengono infine stimate campione per campione risolvendo un problema inverso bayesiano: la verosimiglianza è costruita sulla distanza di Aitchison, la posteriore è campionata con MCMC (DEMetropolisZ) e il modello diretto è sostituito da un surrogato neurale.

Le composizioni osservate sono ricostruite su tutto il dataset, e i parametri stimati restano validi quando si torna dal surrogato al modello completo. Il paziente non è più un dato ma un modello che si può simulare e perturbare: un primo passo verso un digital twin del sistema ospite-microbiota.

Parole chiave: microbiota intestinale, analisi di sensitività globale, indici di Sobol', problema inverso bayesiano, MCMC, modello surrogato neurale

Contents

Abstract **i**

Abstract in lingua italiana **iii**

Contents **v**

Introduction **1**

1 Real Data **5**
1.1 Dataset 5
1.2 Dataset cleaning 7
1.3 Longitudinal data 9

2 Mathematical Model **11**
2.1 Model overview 11
2.2 State variables and notation 12
2.3 Mass balances in the proximal and transverse colon 13
2.4 Mass balances in the distal colon 15
2.5 Reaction network and kinetics 16
2.6 Parameters 19
2.7 Numerical solution of the model 20

3 Model adaptation and analysis **21**
3.1 Matching data with the physical model 21
3.2 The ileal inoculum 24
3.3 The adherence coefficient 25
3.4 Numerical setup and convergence 25
3.5 Robustness to the initial conditions 26

4 Sensitivity analysis **29**

4.1 Introduction . . . 29
4.2 Setup of the analysis . . . 31
4.3 Results and choice of the parameters . . . 33

5 Inverse problem **39**
5.1 Introduction . . . 39
5.2 Setup of the inversion . . . 42
5.3 Surrogate model . . . 43
5.4 Reconstruction of a representative sample . . . 44
5.5 Reconstruction across the dataset . . . 45

6 Conclusions and future developments **47**

Bibliography **49**

A Compositional data **55**
A.1 The simplex . . . 55
A.2 Aitchison geometry . . . 55
A.3 The centred log-ratio transformation . . . 56
A.4 Perturbing a composition . . . 56
A.5 Centre and total variance . . . 58
A.6 Zeros . . . 58

B Model Nomenclature, Parameters and Supporting Material **59**
B.1 Purpose of this appendix . . . 59
B.2 Nomenclature . . . 59
B.3 Numerical values of the parameters . . . 62
B.4 Reaction network, kinetics and stoichiometry . . . 67
B.5 Simulation settings . . . 71

C Numerical implementation **73**
C.1 Structure of the code . . . 73
C.2 Integration . . . 73
C.3 Libraries . . . 74
C.4 Verification . . . 74
C.5 Acceleration with Numba . . . 74

D Sensitivity **77**
D.1 Input factors 77
D.2 Implementation 79
D.3 Second-order Sobol' indices 79

E Inverse problem **81**
E.1 Low-discrepancy design 81
E.2 Training of the surrogate 81
E.3 Check against the full model 82
E.4 Implementation of the sampler 84
E.5 Sampler comparison 84
E.6 Implementation of the cohort run 85

List of Figures **87**

List of Tables **91**

Introduction

Each of us carries about 3.8×10^{13} bacteria in the colon: more than a hundred times the number of stars in the Milky Way, and more than the number of cells in our own body [44]. They are not passengers. They break down the fibres we cannot digest, make vitamins, train the immune system and keep pathogens out [26]. About seventy per cent of the body's immune cells sit in the gut [20], which makes it our largest immune organ. A lot of our neurochemistry happens there too: roughly ninety per cent of the body's serotonin is made in the gut rather than in the brain [26], and the bacteria living there have a real effect on mood, behaviour and thinking [12]. Through the vagus nerve, the immune system and the metabolites it sends into the bloodstream, the colon keeps talking to the rest of the body [26].

Studying it means first of all understanding the diseases that start there: colorectal cancer, the second deadliest tumour in the world [7], and inflammatory bowel disease, whose global number of cases has increased by forty-seven per cent in the last thirty years [49].

But that is only the beginning. An imbalanced gut microbiota has been linked to more than fifty conditions across a dozen organ systems: cancer, autoimmune and neurodegenerative disease, neuropsychiatric and neurodevelopmental disorders, metabolic and cardiovascular disease, liver and kidney disease, skin conditions, allergies and respiratory disease, endocrine and reproductive disorders [12, 50]. Some go as far as to argue that every autoimmune disease begins with an intestine that has turned permeable [13]. The idea itself is old: for two and a half thousand years the saying that all disease begins in the gut has been attributed to Hippocrates.

And yet the same diet does different things to different people. Given identical meals, metabolic responses vary by more than a hundred per cent from one person to the next [5], which is not surprising once you know how much the gut microbiota itself differs between individuals [6]. This is why the evidence on almost every diet ever proposed points in both directions at once.

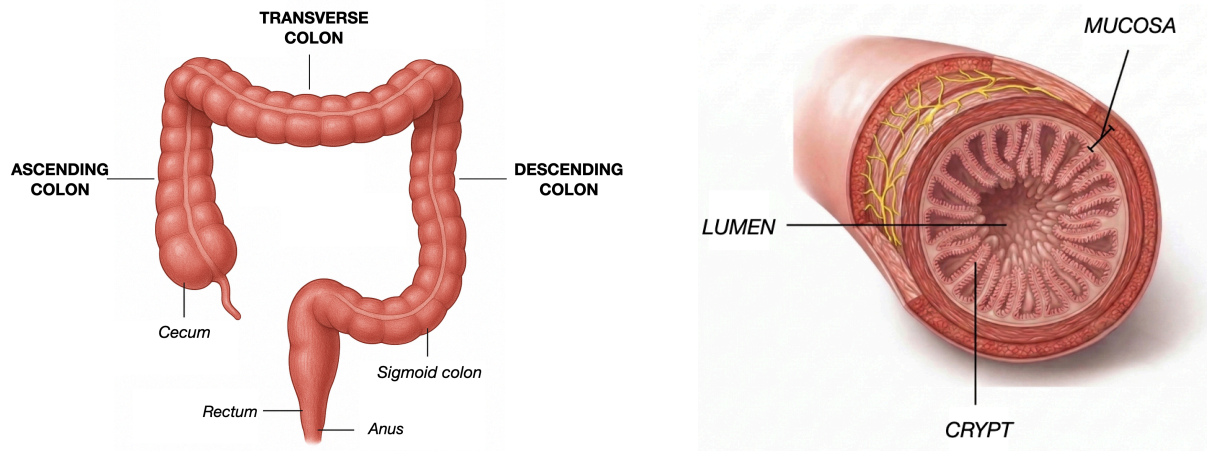
The microbiome is a forest, and many aspects of our lives, from diet and drugs to move-

ment, shape its climate and seasons. But each of us hosts a different forest, and so needs a different climate to do well. Understanding which forest we are looking at, and what grows in it, is a mathematical problem. The goal of this thesis is to make this question accessible through mechanistic modelling and patient-specific inference.

This work was carried out at MaIAGE (INRAE, Université Paris-Saclay) within the Dynenvie team, under the supervision of Béatrice Laroche and Lorenzo Sala, and builds on their long-standing work on the modelling of the intestinal microbiota. The three main contributions that were exploited are the model of carbohydrate degradation of Muñoz-Tamayo et al. [31], the inference of aggregated functional traits of Raguideau et al. [37], and the four functional profiles of Labarthe et al. [23].

Biological background

Food is broken up in the stomach and absorbed in the small intestine. What is left is mostly dietary fibre, which our own enzymes cannot cut, and it reaches the colon almost untouched. There it meets hundreds of species of bacteria, which live without oxygen and ferment it for us. The content moves in one direction only, so the colon is described as three regions in series: proximal, transverse and distal (Figure 1a). Each region is made of two very different environments (Figure 1b): the *lumen*, the content that flows, and the *mucus*, a gel covering the wall in which bacteria can attach without being washed out, and which they can also eat. The fibre is too large to enter a cell, so bacteria first cut it into simple sugars with their own enzymes, a step called *hydrolysis*, and only then ferment these sugars into short chain fatty acids (SCFAs) – acetate, propionate and butyrate – together with lactate and gases; the SCFAs are absorbed by the host and are the products of interest here. Which species do this matters less than one might think, because very different bacteria carry out the same reactions, and these reactions fall into a handful of groups.



(a) Anatomy of the colon.

(b) Cross section of the colonic wall.

Figure 1: The two levels of description used by the model. (a) The ascending, transverse and descending parts, with the sigmoid colon and the rectum, are grouped into three regions in series. (b) Within each region, the lumen carries the flowing content while the wall is covered by a mucus layer that hosts attached bacteria. Adapted from [27].

From DNA to functional profiles

The inside of a working colon is not accessible in practice. Direct sampling requires invasive procedures, so the only routine measurement is a stool sample, that is, what leaves the last region, once. What we read in it is DNA, the molecule that carries the instructions of a cell. The whole DNA of one bacterium is its *genome*, and the genomes of all the microbes living together in a sample form the *metagenome*. A *gene* is a piece of DNA that codes for a protein; when that protein is an *enzyme*, its job is to make one chemical reaction happen fast enough to matter. Reactions run one after the other along a *pathway*, and a group of pathways makes up a *function*. So reading the genes of a stool sample tells us which functions the community is able to carry out, which is not necessarily what it is doing: a gene can be present and never expressed.

This is what shotgun sequencing gives. Older methods read a single gene and only told us which bacteria were there; shotgun reads all the DNA at once, so it tells us both who is present and what they are able to do. Here we only use the second part.

Starting from shotgun data, Labarthe et al. [23] selected the genes involved in the degradation of fibre and mucin and compressed them into four recurring combinations, the *functional profiles*. Every sample is then summarised by four weights (w_1, w_2, w_3, w_4), and this vector is the patient data used in this thesis.

Aim and outline

A single stool sample says what comes out, but nothing about what happens on the way. A mathematical model connects what enters the colon with what leaves it, and gives access to quantities that no measurement can reach.

The aim of this thesis is to reuse the model of carbohydrate degradation proposed in [31] as a quantitative tool, and to understand which of its parameters actually control the outputs that can be measured. Chapter 1 describes the patient data, Chapter 2 the reference model and its numerical implementation. Chapter 3 adapts the model so that its four microbial groups can be matched with the four functional profiles of the data, Chapter 4 identifies the influential parameters through a global sensitivity analysis, and Chapter 5 estimates them for each patient. The conclusions are collected in Chapter 6.

1 | Real Data

1.1. Dataset

The data used in this thesis come from Labarthe et al. [23], who read the DNA of stool samples and turn each one into a short description of what the gut community can do with fibre.

Of the whole gene list they keep only the genes involved in the degradation of fibre and mucin: 33 families of glycoside hydrolases (GH), the enzymes that cut the long chains of fibre into sugars small enough to enter a bacterium, and 91 KEGG orthologies (KO), one for each reaction of the fermentation that follows. A KO is not a single gene but a label: bacteria of different species carry slightly different versions of the same gene, and the label collects all of them, because what matters here is the reaction and not the species doing it. Genes performing the same job are counted together, and jobs that are very close to each other are merged as well, which leaves 101 *aggregated functional traits* (AFTs). Each sample is then a list of 101 numbers, one per metabolic job, saying how much of that job the whole community is able to do.

A constrained non-negative matrix factorisation, following Raguideau et al. [37], compresses these traits into four dimensions. The method builds four fixed combinations of traits, the *functional profiles* (Figure 1.2), and then describes every sample by four weights: the profiles are learned once, and each sample is written as the mixture of them that best reproduces its own 101 frequencies. The weights say how much of each profile a sample carries, so two subjects with similar weights are functionally similar even when their species differ.

The weights cannot be compared between subjects as they are, because their sum also carries the total size of the sample, which depends on how the sample was collected and sequenced. Given that what is comparable is the share that each profile takes inside its own sample, the weights of each sample are divided by their own sum and express them as percentages, so that every sample adds up to 100 %. From here on a sample is described by this composition (w_1, w_2, w_3, w_4); the meaning of the four profiles in terms

of the microbial groups of the model is discussed later, in Section 3.1.

We use the dataset provided by Labarthe et al. [23], containing 3724 faecal metagenomic samples from six international cohorts. The dataset covers healthy subjects and different pathological conditions: inflammatory bowel diseases (Crohn's disease and ulcerative colitis), metabolic diseases such as obesity and type 2 diabetes, and Parkinson's disease. It also includes a cohort studying the effect of a Mediterranean diet, providing a heterogeneous population in terms of health status and gut microbiome composition.

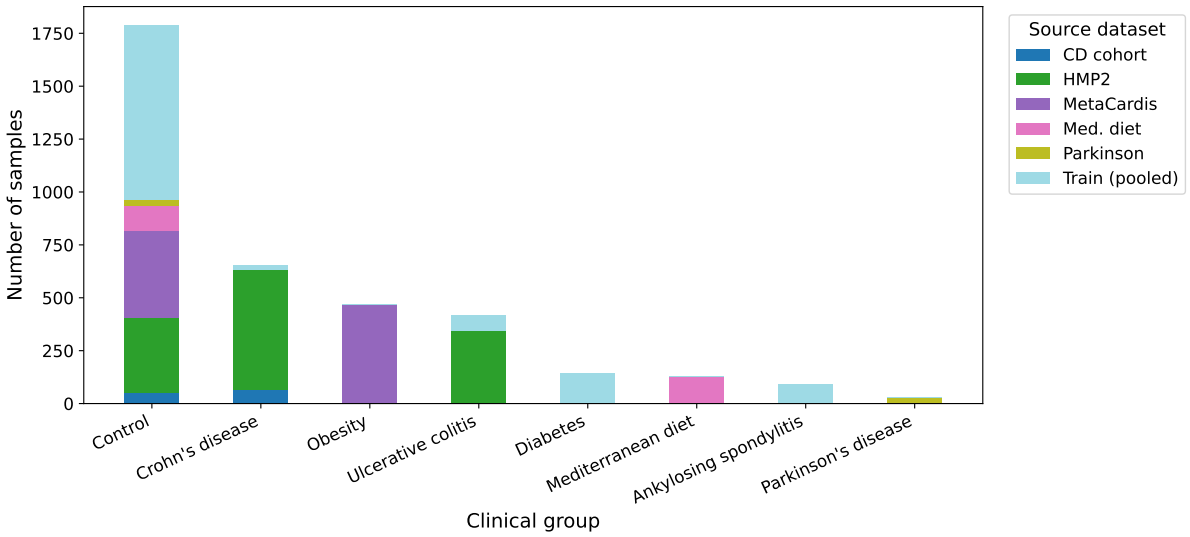


Figure 1.1: Composition of the datasets of Labarthe et al. [23] by clinical group. *Train* is not a cohort but the pooled training set on which the four profiles were inferred (7 studies, 5 countries, balanced in health status); the other five are the external validation cohorts, each selected for one specific effect: *CD cohort*, Crohn's disease; *HMP2*, second Human Microbiome Project (IBD); *MetaCardis*, cardiometabolic cohort; *Med. diet*, Mediterranean-diet intervention; *Parkinson*, Parkinson's disease. The control arm of the Mediterranean-diet trial is counted in *Control*.

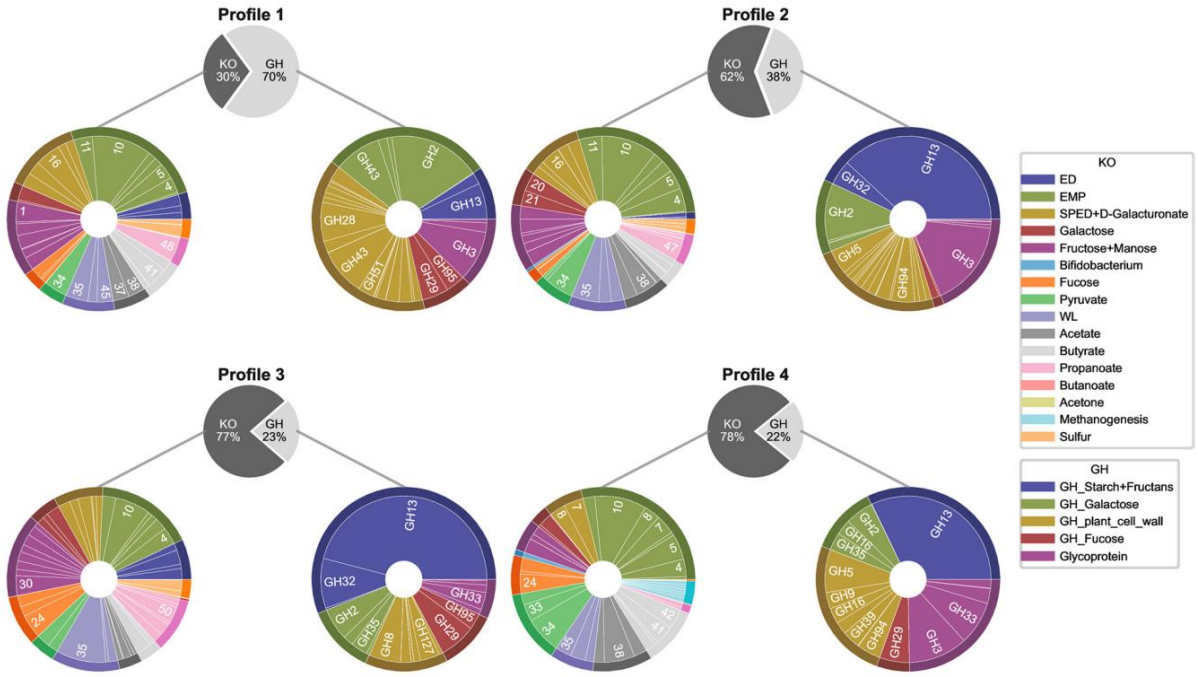


Figure 1.2: Frequency of the 101 AFTs inside each functional profile [23]. Profile 1 is the most hydrolytic, with the largest share of glycoside hydrolases; profiles 2, 3 and 4 are shifted towards the fermentation of simple sugars, and profile 4 is the only one carrying methanogenesis.

1.2. Dataset cleaning

Two points need care before the compositions can be used: weights that are zero, and samples whose composition is extreme.

Absent profiles. The factorisation is constrained to be non-negative, but the solver returns some slightly negative weights. These are zeros up to numerical precision, and they cannot be used as they are: a share cannot be negative, and the log-ratio geometry adopted throughout this work needs strictly positive parts (Appendix A, Section A.6). A weight is therefore informative only if it is clearly above the size of this numerical noise. The largest negative weight in the whole matrix is 3.9×10^{-7}, so we set a threshold at 10^{-6} and treat every weight below it as an absent profile. This is not a rare case: 71.5 % of the samples have at least one absent profile, so they cannot be discarded without losing most of the cohort. We replace such weights by the threshold itself, so that every part stays positive and the percentages stay defined. Since each sample is normalised by its own total, a replaced weight does not map to a single percentage but to a narrow band

of very small values.

The only exception is the first profile. Two isolated samples have w_1 below the threshold, that is a community whose fibre-degrading functions carry no hydrolytic profile at all, which has no biological meaning. These two samples are discarded.

Extreme compositions. A few samples in the upper tail of the three minor profiles are given almost entirely to one of them, up to 99.99 % on w_4. Such compositions have no biological meaning either: a community whose fibre-degrading functions came almost only from the methanogenic profile could not degrade fibre at all. We therefore remove all the samples whose w_2, w_3 or w_4 is in the top 1 %, that is 38 samples for each profile and 109 in total. The first profile is not trimmed.

Together with the two samples above, 111 of 3724 samples are discarded (3.0 %), leaving 3613 samples (Figure 1.3).

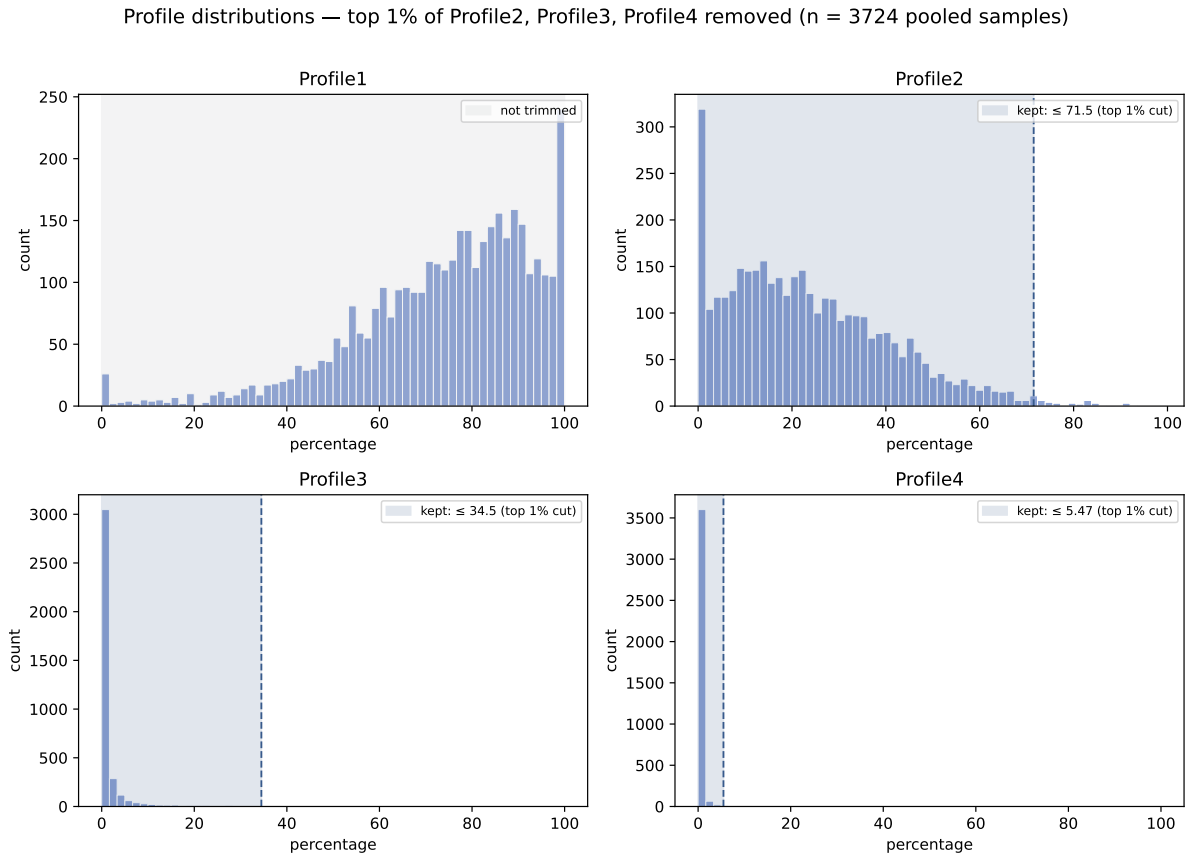


Figure 1.3: Distribution of the four profiles over the pooled cohort, with the retained region shaded.

Profile	absent	min	max
w_1	0	2.4950	99.99
w_2	6.1	0.0417	71.43
w_3	37.4	0.0051	34.48
w_4	54.6	0.0041	5.34

Table 1.1: The cleaned dataset (3613 samples). All values are percentages: the first column gives the share of samples in which the profile is absent, the other two are computed on the remaining samples only.

Table 1.1 gives the range of compositions that we keep. The first profile is present in every sample and dominates most of them, the second one is the main minor profile, while the third and the fourth stay small and are absent in a large part of the cohort. In Chapter 5 we will solve the inverse problem of the model only for these 3613 compositions.

1.3. Longitudinal data

The gut microbiome is not static. Longitudinal data show that its taxonomic composition can change substantially over time, especially in IBD [24]. In the same study, functional measurements showed a less pronounced long-term divergence, supporting the idea that functional organization is more conserved than taxonomy. This is also consistent with Raguideau et al. [37], who argue that different microbial species can replace each other while maintaining similar ecosystem functions.

Therefore, considering the relative functional weights as constant in time is not a rigorous biological assumption, but can be seen as an approximation of a more stable underlying functional state around which the microbiome fluctuates.

2 | Mathematical Model

2.1. Model overview

The model of [31] describes the anaerobic degradation of carbohydrates along the human colon. Its structure is shown in Figure 2.1. As the colon is split into three regions in series, proximal, transverse and distal, and each region into a lumen and a mucus layer, the system has six compartments. Each compartment is treated as a continuous stirred tank reactor (CSTR): a tank that is fed continuously, is drained continuously, and whose content is kept perfectly mixed. Two consequences matter here. First, the content of a compartment is described by concentrations that depend on time only and not on position, so the model is a system of ordinary differential equations. Second, what leaves a compartment has the same composition as what is inside it, so every outgoing term is a flow rate times the concentration in the tank itself.

Every state variable obeys the same balance: what enters, minus what leaves, plus what the reactions produce or consume. Transport follows a fixed pattern. The three lumen compartments are in series and carry the flow coming from the upper digestive tract. Lumen and mucus of the same region exchange material in both directions: soluble components diffuse, bacteria attach to the gel and are torn off by the shear of the flow, and the mucus releases mucins into the lumen. The host closes the system from outside, secreting mucins into the mucus at a rate Γ and absorbing short chain fatty acids and water from both phases. Each compartment is also in contact with a gas phase, which exchanges hydrogen, methane and carbon dioxide with the liquid.

The reactions are the same in all six compartments: hydrolysis of the polysaccharides into sugars, fermentation of the sugars, consumption of lactate and of hydrogen by four microbial groups, and first-order decay of the biomass. They enter the balances through the term $\sum_j Y_{i,j}\,\rho_j(\xi)$, where ρ_j is a reaction rate and $Y_{i,j}$ a yield factor, both discussed in Section 2.5. Symbols, units and numerical values are collected in Appendix B.

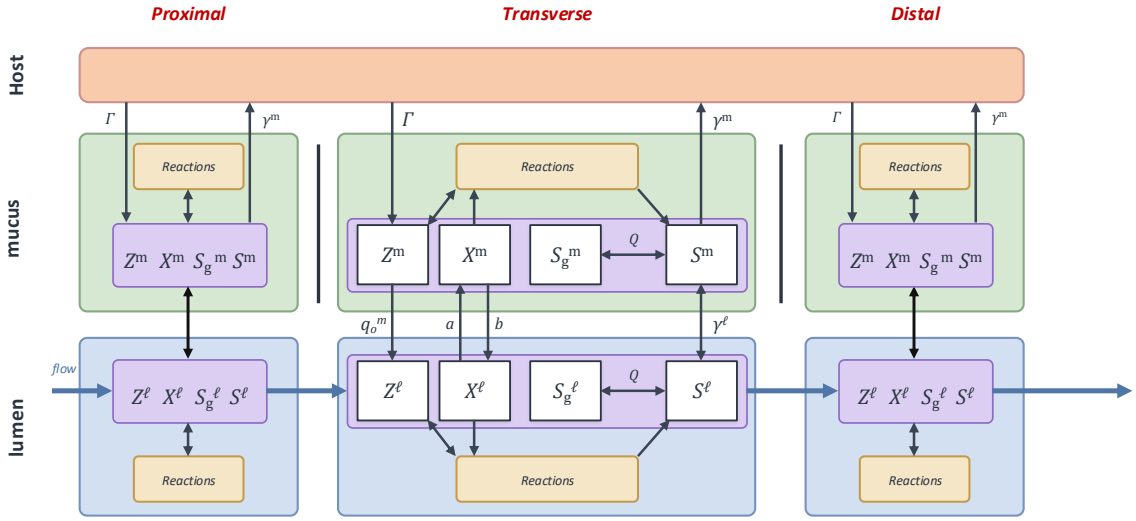


Figure 2.1: Structure of the model. Three regions in series, each split into a lumen and a mucus compartment; the lumen carries the flow from the upper digestive tract to excretion. The transverse region is drawn in detail: in both phases the state variables z, $\mathbf{x}$, $\mathbf{s}$, $\mathbf{s}_g$ are linked by the reactions and by the liquid–gas transfer Q. Lumen and mucus exchange soluble components (γ^l), biomass (adherence a, detachment b) and mucins (q_o^m); the host secretes mucins (Γ) and absorbs metabolites (γ^m).

The complete model is a nonlinear system of 106 coupled ordinary differential equations. Proximal and transverse regions evolve in continuous time with constant volumes, while the distal lumen combines continuous dynamics with discrete excretion events, which reproduce the intermittent nature of defecation.

2.2. State variables and notation

The superscript l refers to the lumen, the superscript m to the mucus. All concentrations are molar (M), except polysaccharides, in g/l; for biomass the empirical molecular weight 113 g/mol is used [4]. In each of the six compartments the state vector is

$$\zeta = \left(\mathbf{s}^{\mathrm{T}},\, z,\, \mathbf{x}^{\mathrm{T}},\, \mathbf{s}_g^{\mathrm{T}}\right)^{\mathrm{T}}, \tag{2.1}$$

where $\mathbf{s} = (s_{\mathrm{su}}, s_{\mathrm{la}}, s_{\mathrm{H_2}}, s_{\mathrm{ac}}, s_{\mathrm{pro}}, s_{\mathrm{bu}}, s_{\mathrm{CH_4}}, s_{\mathrm{CO_2}}, s_{\mathrm{H_2O}})$ are the nine soluble components, in the order glucose, lactate, hydrogen, acetate, propionate, butyrate, methane, carbon

dioxide and water; $\mathbf{x} = (x_{\mathrm{su}}, x_{\mathrm{la}}, x_{\mathrm{H_2a}}, x_{\mathrm{H_2m}})$ are the four microbial groups, each named after the substrate it uses, the last two being homoacetogens and methanogens; and $\mathbf{s}_g = (s_{g,\mathrm{H_2}}, s_{g,\mathrm{CH_4}}, s_{g,\mathrm{CO_2}})$ are the gas-phase concentrations. The scalar z is the concentration of complex polysaccharides: dietary polysaccharides plus mucins coming from the mucus in the lumen, host-secreted mucins in the mucus. The vector ξ collects the liquid-phase concentrations, so that ξ^l and ξ^m refer to the lumen and to the mucus gel.

Following the Petersen matrix convention, components are numbered from 1 to 14: 1–9 are the soluble components in the order above, 10 is z and 11–14 are the microbial groups. Each compartment carries $9 + 1 + 4 + 3 = 17$ states, so the colon accounts for $17 \times 2 \times 3 = 102$ equations; the three equations of the small-intestine cascade and the one for the variable luminal volume V_l of the distal colon give the remaining four.

2.3. Mass balances in the proximal and transverse colon

Proximal and transverse colon have the same structure and constant volumes, so the equations below hold for both. The liquid and gas volumes of the lumen are V_l and V_g, the gel and gas volumes of the mucus are v_m and v_g.

Lumen. The balances of the soluble components, of the polysaccharides, of the microbial groups and of the gases read

$$\dot{s}_i^{\,l} = \frac{q_{\mathrm{in}}}{V_l}\, s_{i,\mathrm{in}}^l - \frac{q_{\mathrm{out}}}{V_l}\, s_i^l - \gamma_i^l s_i^l + \sum_{j=1}^{5} Y_{i,j}\, \rho_j(\xi^l) - Q_i^l, \tag{2.2}$$

$$\dot{z}^{\,l} = \frac{q_{\mathrm{in}}}{V_l}\, z_{\mathrm{in}}^l + \frac{q_{\mathrm{out}}^m}{V_l}\, z^m - \frac{q_{\mathrm{out}}}{V_l}\, z^l - \rho_1(\xi^l), \tag{2.3}$$

$$\dot{x}_i^{\,l} = \frac{q_{\mathrm{in}}}{V_l}\, x_{i,\mathrm{in}}^l - \frac{x_i^l}{\tau_i + V_l/q_{\mathrm{out}}} + b_i \frac{v_m}{V_l}\, x_i^m - a_i x_i^l + \sum_{j=2}^{9} Y_{i,j}\, \rho_j(\xi^l), \tag{2.4}$$

$$\dot{s}_{g,i}^{\,l} = \frac{q_{g,\mathrm{in}}}{V_g}\, s_{g,i,\mathrm{in}}^l - \frac{q_{g,\mathrm{out}}}{V_g}\, s_{g,i}^l + Q_i^l \frac{V_l}{V_g}. \tag{2.5}$$

Here q_{in} and q_{out} are the liquid flow rates entering and leaving the lumen, q_{out}^m is the flow of mucins coming from the mucus and Q_i^l is the liquid–gas transfer rate. The coefficient γ_i^l describes transport towards the host: for SCFA it is an absorption factor and the term is convective, while for sugars the exchange with the mucus is a diffusion, so that $\gamma_i^l s_i^l$ becomes $\gamma_i \left(s_i^l - s_i^m\right)/V_l$. Bacteria leave the compartment with an additional residence time τ_i, which accounts for aggregation, and are exchanged with the mucus through

adherence (a_i) and detachment by shear (b_i).

The effluent flow rate is computed so as to keep V_l constant,

$$q_{\text{out}} = q_{\text{in}} + q^m_{\text{out}} - \sum_{i\in\mathcal{S}} \nu_i\, \gamma^l_i\, s^l_i\, V_l + \sum_{i\in\mathcal{X}} \nu_i \left(b_i\, x^m_i\, v_m - a_i\, x^l_i\, V_l\right), \tag{2.6}$$

where $\nu_i = w_i/\rho_i$ is the molar volume of component i, $\mathcal{S} = \{2,4,5,6,9\}$ are the soluble components absorbed by the host and $\mathcal{X} = \{11,\ldots,14\}$ the microbial groups. Equation (2.6) also imposes conditions on the transport coefficients, needed to keep q_{out} positive. For the proximal colon q_{in} is a constant input from the upper digestive tract; for the transverse colon it equals q_{out} of the proximal lumen. The liquid–gas transfer rate and the outgoing gas flow rate are

$$Q^l_i = k_L a \left(s^l_i - K_{H,i} RT\, s^l_{g,i}\right), \qquad q_{g,\text{out}} = q_{g,\text{in}} + q_{g,\text{lgt}}, \tag{2.7}$$

$$q_{g,\text{lgt}} = \frac{RT\, V_l}{P_{\text{atm}} - p_{\text{H}_2\text{O}}} \left(Q_{\text{H}_2} + Q_{\text{CH}_4} + Q_{\text{CO}_2}\right), \tag{2.8}$$

with $k_L a$ the transfer coefficient times the specific transfer area, K_H the Henry law coefficient, R the ideal gas constant, T the absolute temperature, P_{atm} the atmospheric pressure and $p_{\text{H}_2\text{O}}$ the water vapour pressure. The term $q_{g,\text{lgt}}$ is the gas produced by the transfer from the liquid.

Mucus. In the gel phase the balances are

$$\dot{s}^m_i = \gamma^l_i s^l_i \frac{V_l}{v_m} - \gamma^m_i s^m_i + \sum_{j=1}^{5} Y_{i,j}\, \rho_j(\xi^m) - Q^m_i, \tag{2.9}$$

$$\dot{z}^m = \frac{\Gamma}{v_m} - \frac{q^m_{\text{out}}}{v_m}\, z^m - \rho_1(\xi^m), \tag{2.10}$$

$$\dot{x}^m_i = a_i \frac{V_l}{v_m}\, x^l_i - b_i x^m_i + \sum_{j=2}^{9} Y_{i,j}\, \rho_j(\xi^m), \tag{2.11}$$

and the flow rate from the mucus to the lumen that keeps v_m constant is

$$q^m_{\text{out}} = \frac{\Gamma}{1000 \cdot 0.05} - \sum_{i\in\mathcal{S}} \nu_i\, \gamma^m_i\, s^m_i\, v_m + \sum_{i\in\mathcal{X}} \nu_i \left(a_i\, x^l_i\, V_l - b_i\, x^m_i\, v_m\right). \tag{2.12}$$

The polysaccharides of this compartment are the mucins produced by the host at rate Γ, and the constant dividing Γ in (2.12) is the mucin content of the gel, a density of 1000 g/l times a mucin fraction of 5%. The gas-phase equations are the same as in the lumen,

with V_l replaced by v_m, V_g by v_g and the superscript l by m.

2.4. Mass balances in the distal colon

The liquid content of the distal lumen is not released continuously but excreted in discrete events. Excretion is represented by a valve, closed until the liquid volume reaches a threshold, so that between two events the compartment behaves as a semi-batch reactor: it keeps filling and nothing leaves it. The total volume V_D is constant, but the liquid and gas volumes, with $V_l + V_g = V_D$, are variable and satisfy

$$\dot{V}^{\,l} = -\dot{V}^{\,g} = q_n, \tag{2.13}$$

the net flow rate q_n being the right-hand side of (2.6). The mass balances become

$$\dot{s}_i^{\,l} = \frac{q_{\text{in}}}{V_l}\, s_{i,\text{in}}^l - \frac{q_n}{V_l}\, s_i^l - \gamma_i^l s_i^l + \sum_{j=1}^{5} Y_{i,j}\, \rho_j(\xi^l) - Q_i^l, \tag{2.14}$$

$$\dot{z}^{\,l} = \frac{q_{\text{in}}}{V_l}\, z_{\text{in}}^l + \frac{q_{\text{out}}^m}{V_l}\, z^m - \frac{q_n}{V_l}\, z^l - \rho_1(\xi^l), \tag{2.15}$$

$$\dot{x}_i^{\,l} = \frac{q_{\text{in}}}{V_l}\, x_{i,\text{in}}^l - \frac{q_n}{V_l}\, x_i^l + b_i \frac{v_m}{V_l}\, x_i^m - a_i x_i^l + \sum_{j=2}^{9} Y_{i,j}\, \rho_j(\xi^l), \tag{2.16}$$

$$\dot{s}_{g,i}^{\,l} = \frac{q_{g,\text{in}}^l}{V_g}\left(s_{g,i,\text{in}}^l - s_{g,i}^l\right) - \frac{q_{g,\text{lgt}}^l - q_n}{V_g}\, s_{g,i}^l + Q_i^l \frac{V_l}{V_g}, \tag{2.17}$$

where q_{in} equals q_{out} of the transverse lumen. The mucus equations keep the structure of the previous ones.

The distal lumen is a hybrid system: the continuous dynamics (2.13)–(2.17) is interrupted by a discrete reset. When the volume occupied by the chyme reaches the threshold $V_{\max}$, a bowel movement occurs and the liquid volume is reset instantaneously,

$$V_l(t^-) = V_{\max} \;\longrightarrow\; V_l(t^+) = V_{\text{reset}}, \qquad V_{\text{reset}} < V_{\max}, \tag{2.18}$$

with numerical values given in Appendix B. Concentrations do not change at the reset, because the excreted material has the same composition as the retained one.

Input from the small intestine. The upper digestive tract is not modelled. Its only role here is to deliver the polysaccharides that escaped digestion and to spread the peaks of the meals over time. This is done with three linear compartments in series. They act

as a low-pass filter: the fast changes of the diet are smoothed out, while the total amount delivered is preserved.

$$\dot{z}_j = D\,(z_{j-1} - z_j)\,, \qquad j = 1, 2, 3, \tag{2.19}$$

with D the dilution rate, z_0 the diet and z_3 the input of the proximal lumen.

2.5. Reaction network and kinetics

The reaction network. The biochemical core of the model is shown in Figure 2.2: hydrolysis of the polysaccharides z into sugars s_{su}, fermentation of the sugars into short chain fatty acids, lactate and gases, and the two closing steps of lactate and hydrogen consumption. In the mass balances the network is condensed into nine processes: hydrolysis ($j = 1$), uptake of glucose, lactate and hydrogen by the four microbial groups ($j = 2, \ldots, 5$) and decay of the same groups ($j = 6, \ldots, 9$). Each process carries a rate $\rho_j(\xi)$ and a set of yield factors $Y_{i,j}$, the amount of component i produced, or consumed if negative, per unit of reaction. The twelve chemical reactions behind the arrows and the two Petersen matrices are collected in Appendix B.4.

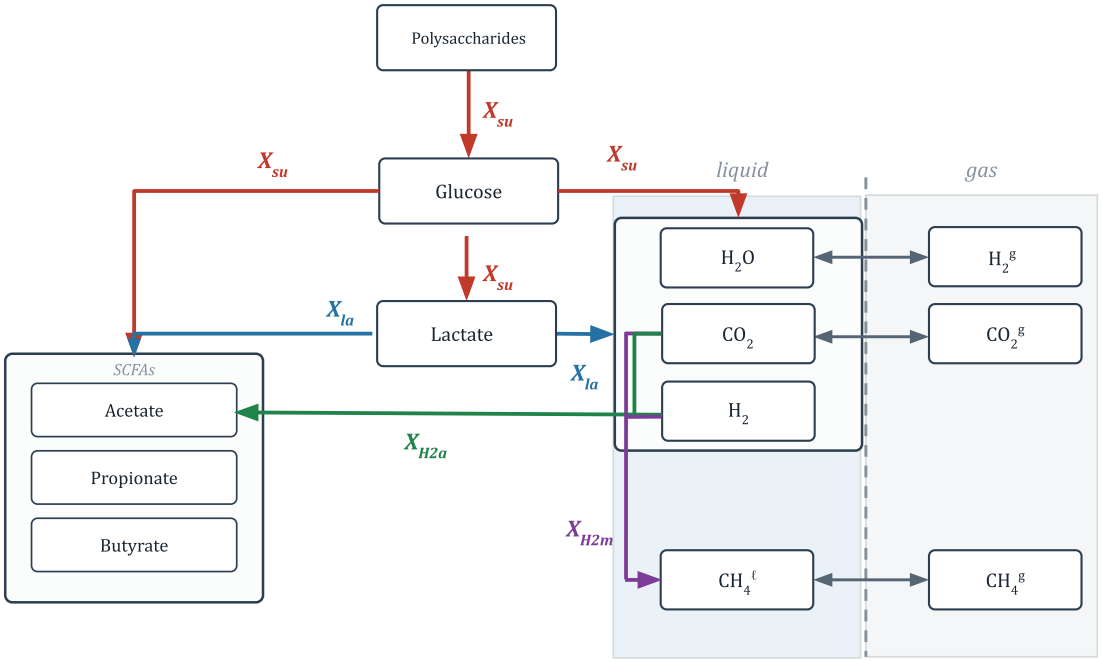


Figure 2.2: Aggregated metabolic pathway, the graphical counterpart of the reactions of Table B.12 (Appendix B.4). Each arrow is labelled by the microbial group that catalyses it: x_{su} hydrolyses polysaccharides into glucose and ferments glucose into lactate, short chain fatty acids and gases; x_{la} turns lactate into short chain fatty acids and gases; $x_{\mathrm{H_2a}}$ and $x_{\mathrm{H_2m}}$ consume hydrogen and produce acetate and methane respectively. Once dissolved, H_2, CH_4 and CO_2 also move between the liquid and the gas phase of the compartment.

Process rates. The first five processes are catalysed by bacteria, so their rates are proportional to the concentration of the group that carries them out; this group is the first argument of ρ_j. Hydrolysis follows a Contois law, which depends on the substrate-to-biomass ratio; the four uptake processes follow Monod kinetics, with maximum rate $k_{m,i}$ and half-saturation constant $K_{s,i}$; decay is first order, with the same constant k_d for the four groups. Methanogenesis is multiplied by an empirical inhibition function I_{pH},

the only pH effect kept in the model. The nine rates read

$$\begin{aligned}
\rho_1(x_{\mathrm{su}}, z) &= k_{\mathrm{hyd},z}\,\frac{z\,x_{\mathrm{su}}}{K_{x,z}x_{\mathrm{su}} + z}, & \rho_6(x_{\mathrm{su}}) &= k_d\,x_{\mathrm{su}},\\
\rho_2(x_{\mathrm{su}}, s_{\mathrm{su}}) &= k_{m,\mathrm{su}}\,\frac{s_{\mathrm{su}}\,x_{\mathrm{su}}}{K_{s,\mathrm{su}} + s_{\mathrm{su}}}, & \rho_7(x_{\mathrm{la}}) &= k_d\,x_{\mathrm{la}},\\
\rho_3(x_{\mathrm{la}}, s_{\mathrm{la}}) &= k_{m,\mathrm{la}}\,\frac{s_{\mathrm{la}}\,x_{\mathrm{la}}}{K_{s,\mathrm{la}} + s_{\mathrm{la}}}, & \rho_8(x_{\mathrm{H_2a}}) &= k_d\,x_{\mathrm{H_2a}},\\
\rho_4(x_{\mathrm{H_2a}}, s_{\mathrm{H_2}}) &= k_{m,\mathrm{H_2a}}\,\frac{s_{\mathrm{H_2}}\,x_{\mathrm{H_2a}}}{K_{s,\mathrm{H_2a}} + s_{\mathrm{H_2}}}, & \rho_9(x_{\mathrm{H_2m}}) &= k_d\,x_{\mathrm{H_2m}},\\
\rho_5(x_{\mathrm{H_2m}}, s_{\mathrm{H_2}}) &= k_{m,\mathrm{H_2m}}\,\frac{s_{\mathrm{H_2}}\,x_{\mathrm{H_2m}}}{K_{s,\mathrm{H_2m}} + s_{\mathrm{H_2}}}\,I_{\mathrm{pH}},
\end{aligned} \tag{2.20}$$

with I_{pH} specified in the Appendix B.4. Here each process is catalysed by a single group, as in [31]; the first argument is kept explicit because in Chapter 3 the biomass entering each process is redefined.

The fractions f and η. The rates say how fast each process runs; the yields $Y_{i,j}$ say what it produces. They are not free parameters, because they follow from the chemistry of the reactions. Each arrow of Figure 2.2 does two things at once. Part of the substrate is broken down into metabolites and provides energy: this is *catabolism*, and it is what the arrow shows. The remaining part is converted into new biomass: this is *anabolism*, it does not appear in the figure, and it is the reason why the group grows. The parameter $f \in [0, 1]$ is the fraction of substrate sent to catabolism, so $1 - f$ is the fraction sent to growth. When catabolism can follow more than one route, a second set of fractions $\eta_k \in [0, 1]$, with $\sum_k \eta_k = 1$, says how the catabolic flux splits among the routes.

Lactate utilization is the simplest example (Figure 2.3). A lactate molecule taken up by x_{la} is either degraded into propionate and acetate, or degraded into butyrate, or used to build biomass. One fraction f_{la} and one fraction $\eta_{\mathrm{la},1}$ are therefore enough. Glucose works in the same way with four catabolic routes, which give lactate, acetate, propionate and butyrate, hence three independent η; homoacetogens and methanogens have a single route each, so for them f alone is enough.

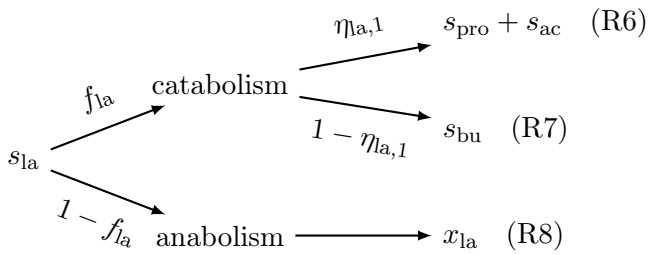


Figure 2.3: The single arrow "lactate utilization" of Figure 2.2, split into its catabolic and anabolic parts. The fraction f_{la} of the lactate taken up is degraded, the fraction $1 - f_{\mathrm{la}}$ becomes biomass, and $\eta_{\mathrm{la},1}$ splits the catabolic flux between the two degradation routes. Gases and water are omitted; R6–R8 are the reactions of Table B.12.

Once f and the η_k are given, every yield $Y_{i,j}$ of the process follows from the reactions by elementary mass and elemental balances; the derivation is carried out in Appendix B.4. Two consequences are used repeatedly in the following chapters. First, $1-f$ *is* the growth yield of the group, up to a constant: a larger f means more metabolites and less biomass per unit of substrate. Second, the η_k do not change how much is produced in total, but only *what* is produced.

Reparametrization. The yield factors of the four microbial processes can be replaced by the eight independent quantities

$$\theta_{\mathrm{stoich}} = (f_{\mathrm{su}},\, \eta_1,\, \eta_2,\, \eta_3,\, f_{\mathrm{la}},\, \eta_{\mathrm{la},1},\, f_{\mathrm{H_2a}},\, f_{\mathrm{H_2m}}) \in [0,1]^8, \qquad \eta_1 + \eta_2 + \eta_3 \leq 1. \tag{2.21}$$

In this work the model is implemented directly in terms of θ_{stoich}: the yields are recomputed from (2.21) at every parameter change, instead of being stored as fixed numbers. Every parameter set explored in the following chapters is then chemically consistent by construction, and the quantities that are varied have a clear meaning: how much substrate goes to growth rather than to metabolites, and how the fermentation is oriented towards lactate, acetate, propionate or butyrate. Perturbing the individual yields one at a time would break the balances and make the sensitivity analysis of Chapter 4 meaningless.

2.6. Parameters

Besides θ_{stoich}, the model uses kinetic parameters ($k_{\mathrm{hyd},z}$, $K_{x,z}$, $k_{m,i}$, $K_{s,i}$, k_d, pH_{L}, pH_{U}), transport parameters (γ_i^l, γ_i^m, a_i, b_i, τ_i, k_La, $K_{H,i}$) and physiological ones (V_l, V_g, v_m, v_g, q_{in}, Γ, T, P_{atm}, $p_{\mathrm{H_2O}}$). Written in full, the 102 colonic equations would involve 333 coefficients; [31] brings them down to 60 by giving most of them the same value in all

compartments. Symbols and units are listed in Appendix B.2, baseline values in Appendix B.3. Which parameters really matter for the outputs of interest provided by the global sensitivity analysis of Chapter 4.

2.7. Numerical solution of the model

The model of [31] was originally implemented in `Matlab/Simulink`. I have entirely re-implemented it in `Python`, in order to couple it with the sampling, optimization and statistical libraries needed by the next chapters; the implementation details are collected in Appendix C. The system (2.2)–(2.19) is written in the compact form $\dot{\mathbf{y}} = f(\mathbf{y}, \theta, t)$, with $\mathbf{y} \in \mathbb{R}^{106}$ and θ collecting all the parameters, the stoichiometric block θ_{stoich} included.

Fast reaction kinetics and slow transport make the system stiff, so an implicit integrator is required: time integration uses `scipy.integrate.solve_ivp` with the `BDF` method, the counterpart of the `ode15s` solver of the original implementation. The hybrid dynamics of the distal colon is not left to the integrator: the crossing $V_l = V_{\max}$ is located by event detection, the integration is stopped, the state is reset and the integration is restarted. Since the following chapters require thousands of evaluations of the right-hand side, it has been compiled with `Numba`, which gives a speed-up of about a factor of seven per simulation. Under identical parameters, initial conditions and inputs, the implementation reproduces the results of [31].

Final remarks. This formulation is the mechanistic basis of the thesis: a nonlinear compartmental system with mixed continuous–discrete dynamics, whose stoichiometry ties the yield factors to a small set of interpretable parameters. The next chapters revisit and adapt it to the objectives of the thesis.

3 | Model adaptation and analysis

The model of Chapter 2 describes a generic colon, not the system host-microbiota of a given person. Three of its choices have to be changed before it can be compared with the patient data presented in Chapter 1:

- the four microbial states, each associated to a single reaction, are replaced by populations that can carry several reactions, so that they match the four functional profiles of the data (Section 3.1);
- the bacterial inflow from the ileum, which fixes the lowest abundance the model can reach, is recalibrated (Section 3.2);
- the adherence coefficient, set to zero in the reference model, is given a small positive value, so that the mucus is continuously reseeded from the lumen (Section 3.3).

The rest of the chapter explains the initial state and the rule used to stop a run (Section 3.4), then the effect of the initial conditions on the stationary regime (Section 3.5).

From here on, "the model" always means this adapted version, and it is the one used in all the following chapters.

3.1. Matching data with the physical model

In Chapter 1 every sample is described by a weight for each of the four functional profiles, summarised in Figure 1.2. As discussed in Section 1.3, the profile composition of one patient is stable enough over the time scale considered here to be used as a target. The physical model of Chapter 2 reduces the gut microbiota to four biomass states. The comparison can only be made on *compositions*: the weights $\mathbf{w}$ come from a factorisation of gene counts, they are not concentrations, and their absolute values share no unit with the biomasses, which are in g/L. Hence we ask

$$\frac{\mathbf{x}_D^l(t_e)}{\|\mathbf{x}_D^l(t_e)\|_1} \approx \frac{\mathbf{w}}{\|\mathbf{w}\|_1}, \tag{3.1}$$

where $\mathbf{x}_D^l(t_e)$ collects the four biomasses of the distal lumen at an excretion time t_e, the compartment that becomes faecal material.

It is important to note that the weights are built by counting genes that are present, not genes that are working. Writing (3.1), we assume that every gene expresses its function, so that potential and activity are proportional. Labarthe et al. [23] gives support for this, since the same four profiles are recovered from metatranscriptomic data.

From fixed substrate roles to generalist populations. Equation (3.1) requires each profile to be associated with one state of the model, and this is where the reference formulation is too rigid. There each biomass state is defined by the substrate it uses: x_{su} for sugar utilization, x_{la} for lactate utilization, $x_{\mathrm{H_2a}}$ for homoacetogenesis and $x_{\mathrm{H_2m}}$ for methanogenesis, with x_{su} also driving hydrolysis. The profiles do not follow this rule. Only two of them are close to a single function: Profile 1, made for the most part of glycoside hydrolases, plays the role of x_{su}, and Profile 4, the only one carrying methanogenesis, that of $x_{\mathrm{H_2m}}$. Profiles 2 and 3 are broader and appear in several pathways at once, Profile 2 mainly in the sugar and pyruvate–lactate routes, Profile 3 in propionate production from lactate and in reductive acetogenesis.

Guided by this reading of the profiles, we let the two central states take part in more than one reaction, distributing their activity over the reactions already present in the model: the second state over glucose utilization, lactate utilization and, with a smaller weight, homoacetogenesis, with weights $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3)$ in this order; the third state over the two reactions left for it, lactate utilization and homoacetogenesis, with weights $\boldsymbol{\beta} = (\beta_1, \beta_2)$. Both sets are non-negative and sum to one. The profiles give the order of these contributions, but not their exact size, so we fix them by eye together with the biologists,

$$\boldsymbol{\alpha} = (0.5,\, 0.3,\, 0.2), \qquad \boldsymbol{\beta} = (0.8,\, 0.2), \tag{3.2}$$

the same for all patients: the variability between patients is left to the model parameters estimated later. How much the results depend on this choice is studied in Chapter 4. Since the subscripts su, la, $\mathrm{H_2a}$, $\mathrm{H_2m}$ no longer describe what each group does, from here on we write $\mathbf{x} = (x_1, x_2, x_3, x_4)^{\mathrm{T}}$.

Effective reaction rates. The kinetic laws are not modified. Writing χ for the biomass argument of Eq. (2.20), the five reactive processes are

$$\begin{aligned}
\rho_1(\chi, z) &= k_{\mathrm{hyd},z}\,\frac{z\,\chi}{K_{x,z}\chi + z}, & \rho_2(\chi, s_{\mathrm{su}}) &= k_{m,\mathrm{su}}\,\frac{s_{\mathrm{su}}\,\chi}{K_{s,\mathrm{su}} + s_{\mathrm{su}}},\\
\rho_3(\chi, s_{\mathrm{la}}) &= k_{m,\mathrm{la}}\,\frac{s_{\mathrm{la}}\,\chi}{K_{s,\mathrm{la}} + s_{\mathrm{la}}}, & \rho_4(\chi, s_{\mathrm{H_2}}) &= k_{m,\mathrm{H_2a}}\,\frac{s_{\mathrm{H_2}}\,\chi}{K_{s,\mathrm{H_2a}} + s_{\mathrm{H_2}}},\\
\rho_5(\chi, s_{\mathrm{H_2}}) &= k_{m,\mathrm{H_2m}}\,\frac{s_{\mathrm{H_2}}\,\chi}{K_{s,\mathrm{H_2m}} + s_{\mathrm{H_2}}}\,I_{\mathrm{pH}}.
\end{aligned} \tag{3.3}$$

The four uptake processes (from ρ_2 to ρ_5) are linear in χ, so each of them can be shared among the states that carry it; hydrolysis (ρ_1) is a Contois law and is not linear, but it stays with x_1 alone and never has to be split. Within one compartment the substrate argument is the same for every state, so from here on we leave it implicit and write $\rho_j(\chi)$.

For a soluble component a reaction sees all the biomass that can take part in it, each state entering with its own weight and the specialists x_1 and x_4 with weight one:

$$\begin{aligned}
\widetilde{\rho}_1 &= \rho_1(x_1), & \widetilde{\rho}_2 &= \rho_2(x_1 + \alpha_1 x_2),\\
\widetilde{\rho}_3 &= \rho_3(\alpha_2 x_2 + \beta_1 x_3), & \widetilde{\rho}_4 &= \rho_4(\alpha_3 x_2 + \beta_2 x_3),\\
\widetilde{\rho}_5 &= \rho_5(x_4).
\end{aligned} \tag{3.4}$$

In the balances of Chapter 2 it is then enough to replace ρ_j by $\widetilde{\rho}_j$, so that in any soluble component the reaction term becomes

$$\dot{s}_i|_{\mathrm{reac}} = \sum_{j=1}^{5} Y_{i,j}\,\widetilde{\rho}_j, \qquad \dot{z}|_{\mathrm{reac}} = -\widetilde{\rho}_1. \tag{3.5}$$

Flow, absorption and lumen–mucus exchange are unchanged.

Biomass growth. The biomass equations are different, because each state grows only on the share of the reaction that it carries, obtained by expanding the arguments of (3.4):

$$\begin{aligned}
\dot{x}_1|_{\mathrm{growth}} &= Y_{\mathrm{su}}\,\rho_2(x_1),\\
\dot{x}_2|_{\mathrm{growth}} &= \alpha_1 Y_{\mathrm{su}}\,\rho_2(x_2) + \alpha_2 Y_{\mathrm{la}}\,\rho_3(x_2) + \alpha_3 Y_{\mathrm{H_2a}}\,\rho_4(x_2),\\
\dot{x}_3|_{\mathrm{growth}} &= \beta_1 Y_{\mathrm{la}}\,\rho_3(x_3) + \beta_2 Y_{\mathrm{H_2a}}\,\rho_4(x_3),\\
\dot{x}_4|_{\mathrm{growth}} &= Y_{\mathrm{H_2m}}\,\rho_5(x_4),
\end{aligned} \tag{3.6}$$

where Y_{su}, Y_{la}, $Y_{\mathrm{H_2a}}$ and $Y_{\mathrm{H_2m}}$ are the biomass yields of processes 2 to 5, that is the entries $Y_{11,2}$, $Y_{12,3}$, $Y_{13,4}$ and $Y_{14,5}$ of the Petersen matrix. Decay keeps its reference form, $-k_d x_k$

on each state, and hydrolysis produces no biomass. The reference model is recovered for $\boldsymbol{\alpha} = (0, 1, 0)$ and $\boldsymbol{\beta} = (0, 1)$: each process is then carried by a single state, and (3.4) and (3.6) reduce to the equations of Muñoz-Tamayo et al. [31]. This limit is used as a test of the implementation.

3.2. The ileal inoculum

The term $q_{\text{in}} x^l_{i,\text{in}}/V_l$ of Eq. (2.4) feeds the proximal lumen with bacteria coming from the ileum. In the reference implementation the four inflow concentrations were simply set equal to the initial conditions of the model [31], a composition 60/20/10/10 for a total of 0.1 g/L. Those numbers described the four specialist groups of the original formulation. After Section 3.1 the four states are different populations, so the inflow has to be set again.

The values adopted. Together with the biologists we set the total inflow to $x_{\text{tot}} = 0.01\,\text{g/L}$, the low end of the range obtained from the bacterial counts of ileal effluent [6], and we give almost all of it to the first group: the ileum is dominated by sugar fermenters, while the groups living on lactate and hydrogen are negligible there,

$$\left(x_{1,\text{in}}, x_{2,\text{in}}, x_{3,\text{in}}, x_{4,\text{in}}\right) = \left(0.997,\ 10^{-3},\ 10^{-3},\ 10^{-3}\right) x_{\text{tot}}. \tag{3.7}$$

Why this step is needed. Take one group and sum its balances (2.4), (2.11) and (2.16) over the six compartments, each one multiplied by its own volume, V_l for a lumen and v_m for a mucus. Adherence and shear then cancel in pairs, and what is left is the inflow from the ileum, the growth and the decay of the group, and the material leaving with the stool. Growth is never negative; decay is first order and removes $k_d x$. With the baseline parameters of Appendix B.3 the specific growth rate is orders of magnitude larger than k_d, so the biological term is positive and the biomass increases from the proximal to the distal region, as it does in the gut. What enters is therefore not destroyed on the way, and each group stays above the level that transport alone sustains,

$$x_D \ \geq\ c\, x_{\text{in}}, \qquad c > 0, \tag{3.8}$$

where c depends only on the flow rates, the volumes and k_d. Provided that k_d is not pushed to values with no biological meaning, no choice of the kinetic parameters can bring a group below this floor: the inoculum fixes the smallest abundance the model can represent.

Under (3.1) the floor must lie below the smallest compositions we want to explain, that is 2.5 % for the first profile and about 0.004 % for the smallest of the other three (Table 1.1), the band of the absent profiles lying lower still. With (3.7) the floors are 0.06 % of the distal biomass for the first group and 6×10^{-5} % for the other three, that is 40 to 70 times below those two limits, which leaves room for growth. The reference inoculum is ten times larger and far less skewed, and it would keep the three minor groups above the small compositions of the cohort, which could then not be reproduced at all. Apart from this, the exact values are not critical: changing x_{tot} by a factor of ten does not move the distal steady state, which at stationarity is fixed by growth and not by seeding.

3.3. The adherence coefficient

The third adaptation concerns the exchange of microbial biomass between lumen and mucus. The model uses a_i for adherence from lumen to mucus and b_i for shear from mucus to lumen. Muñoz-Tamayo et al. [31] set $a_i = 0$ and $b_i = 0.08\,\mathrm{d}^{-1}$, so bacteria can leave the mucus but cannot enter it. We keep b_i unchanged and set

$$a_i = 0.01\,\mathrm{d}^{-1}. \tag{3.9}$$

The reason is structural. With $a_i = 0$ the face $\{x_i^m = 0\}$ is invariant: if one mucus population becomes very small, the lumen cannot restore it. A small positive a_i continuously reseeds the mucus and reduces the memory of the initial condition. We keep $a_i \ll b_i$, so shear remains the dominant exchange direction.

3.4. Numerical setup and convergence

The three changes above define the model used in the rest of the thesis. Before using it, we look at how it behaves in practice: a run has to start from a reasonable state and has to be stopped once the stationary regime is reached.

Initial state. Every run needs an initial state. We follow Muñoz-Tamayo et al. [31], who build theirs with a long warm-up under a constant feed of 10 g of fibre per day, and we repeat that computation with the adapted model. The final state of all compartments is then the common initial condition of every simulation in this work: the warm-up is done once, at the baseline parameter set, and is not repeated when a parameter changes. A single starting state is legitimate only because the stationary regime keeps no memory of it, which is what Section 3.5 shows.

Stopping rule. Each parameter set moves away from that common state and can need a different settling time, so runs are stopped by an explicit criterion rather than at a fixed final time. The criterion is written on the four biomasses of the distal lumen sampled at excretion, that is on the model output used in (3.1). The natural choice, stopping as soon as two consecutive excretions return the same value, measures how fast the state is moving at that moment and not how far it still has to go: a system that settles slowly moves little between two excretions because it is slow, not because it has arrived. For this reason, we sum the absolute changes over a time window W:

$$V_n = \max_j \frac{1}{|x_n^{(j)}|} \sum_{i=1}^{k} \left| x_{n-k+i}^{(j)} - x_{n-k+i-1}^{(j)} \right|, \qquad t_n - t_{n-k} \geq W, \tag{3.10}$$

where $x^{(j)}$, $j = 1, \ldots, 4$, are the four distal biomasses. We stop when $V_n < \varepsilon$, with $W = 30$ d and $\varepsilon = 10^{-3}$. Using a time window rather than a fixed number of excretions also keeps the rule usable when the excretion frequency changes.

Figure 3.1 shows the criterion on a run of the adapted model defined above, at the baseline parameter set and started from the warm-up state. The maximum over j lets the slowest group set the stopping time, while the normalisation by $x_n^{(j)}$ puts the four groups on the same relative scale. A strictly constant state is never reached: the meals are imposed as a periodic input whose period is not a multiple of the excretion period, so each excretion samples the feeding cycle at a different phase and the recorded values keep oscillating by a small amount. This residual jitter is why V_n flattens around $2 \cdot 10^{-4}$ instead of going to zero.

3.5. Robustness to the initial conditions

We now ask whether the long-term behaviour depends on the initial state. The best way to answer would be analytically: find the equilibria, check that the interior one is unique, study its stability. This is out of reach here. The model has 106 state variables, the compartments are coupled in series through the flows and in parallel through the lumen–mucus exchange, and the excretion event makes the distal compartment a hybrid system. The object to study is then not a fixed point of a vector field but a fixed point of the return map between two excretions, and that map cannot be written in closed form. We replace the proof by a numerical exploration: every state variable is perturbed around its baseline value, run each initial condition to its stationary regime with the criterion (3.10), and take as output the four biomasses of the distal lumen at the last excretion.

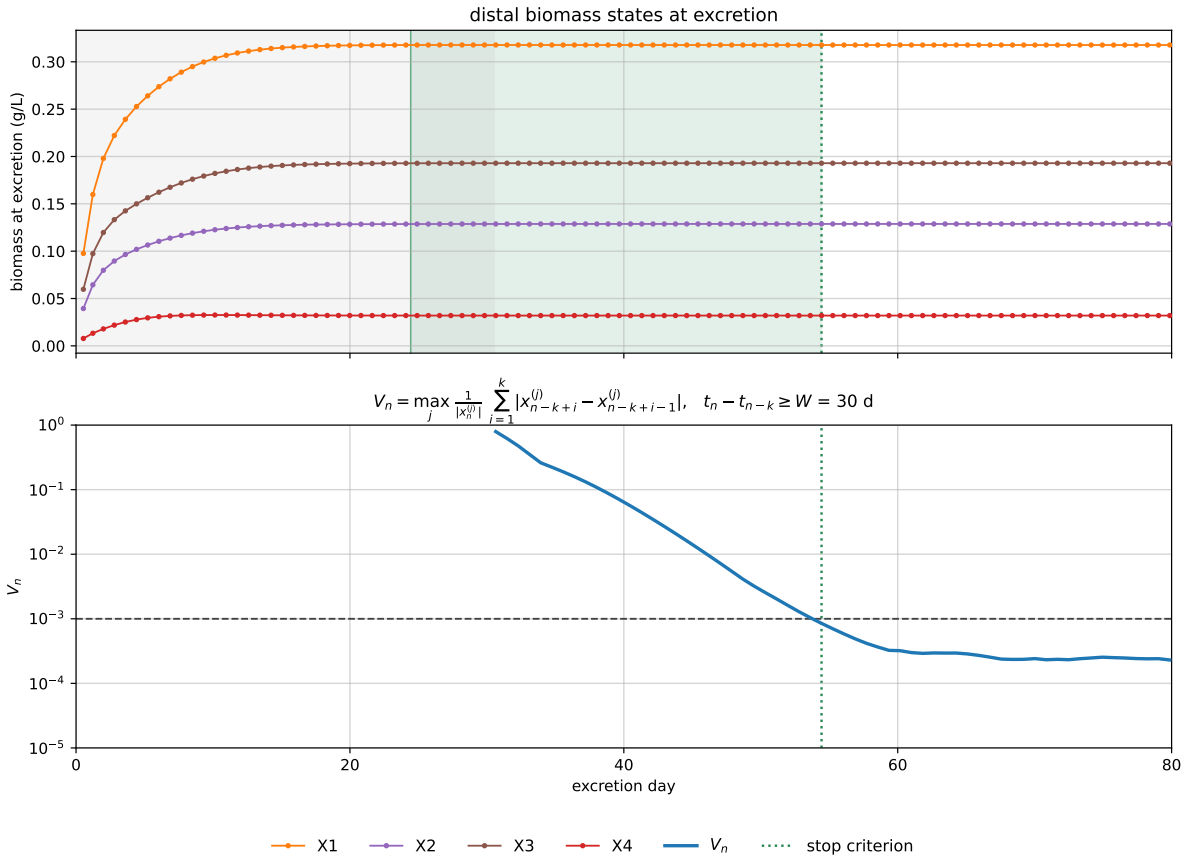


Figure 3.1: Top: the four distal biomasses $x^{(j)}$ recorded at each excretion, in g/L. Bottom: the windowed total variation V_n of Eq. (3.10), in logarithmic scale, against the threshold $\varepsilon = 10^{-3}$ (horizontal dashed line). Since V_n sums the changes over a window of length $W = 30$ d, it is defined only from day 30 on; the green dotted line marks the excretion at which it falls below ε.

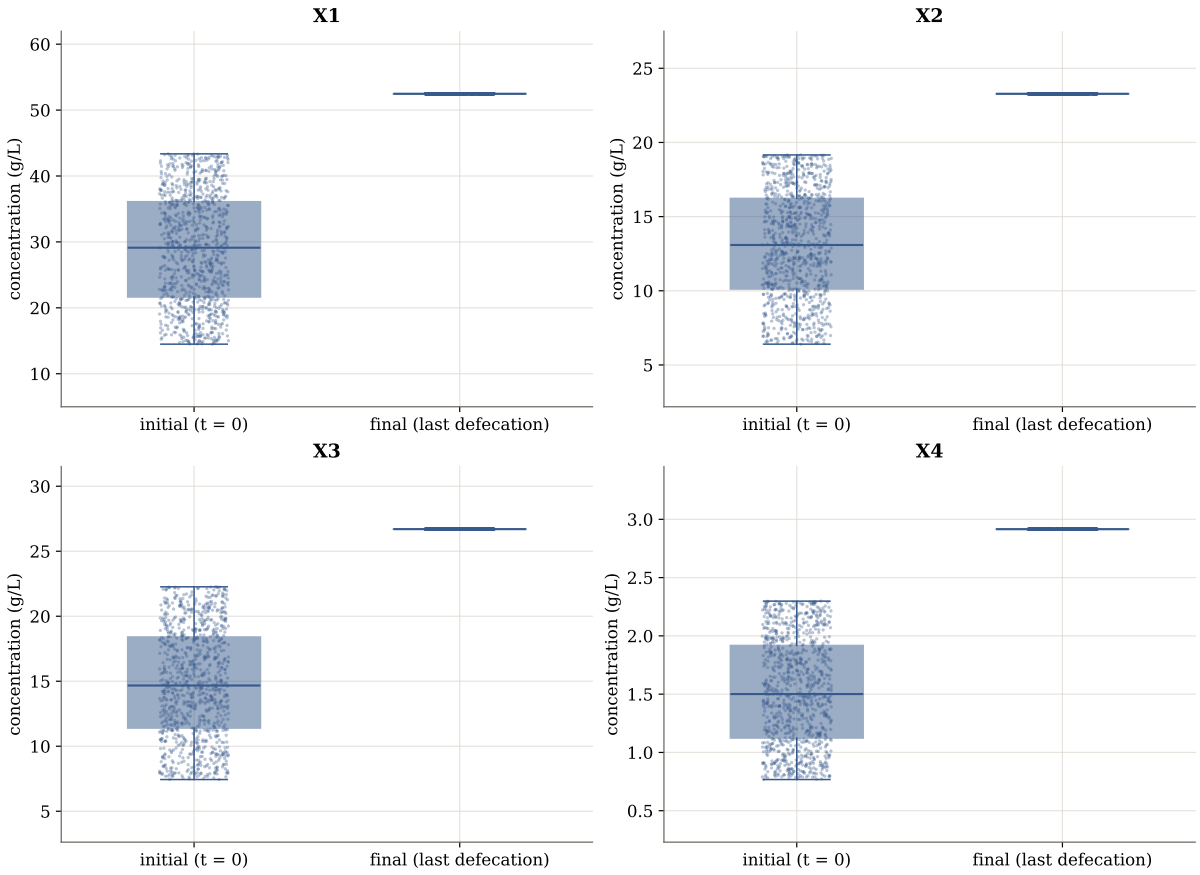


Figure 3.2: Distal biomass of the four groups, in g/L, for $n = 1000$ initial conditions. Each of the 106 state variables is multiplied by an independent factor drawn uniformly in $[1 - \ell, 1 + \ell]$ with $\ell = 0.5$, that is a perturbation of $\pm 50\%$ around the baseline value; all states are perturbed, although only the biomasses are shown here. In each panel the left box is the distribution of the distal biomass imposed at $t = 0$, with the individual samples shown as points, and the right box is the distribution obtained at the last excretion. The right boxes are collapsed to a line at the scale of the figure.

The initial distributions are wide, as imposed, while the final ones collapse onto a single value: for each of the four groups the spread is reduced by about three orders of magnitude. On the relative composition the residual standard deviation is smaller by about five orders of magnitude, which is compatible with the jitter of sampling at the excretion time. Within the sampled region the perturbation is forgotten and the stationary composition is fixed by the parameters alone. As said above, this is an exploration and not a proof, but it is the evidence the following chapters rely on.

4 | Sensitivity analysis

4.1. Introduction

The model contains many parameters, but only some of them can strongly change the distal microbial composition. Before solving the inverse problem, we use global sensitivity analysis to find them [36, 42]. The model is a deterministic map from k input factors $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_k)$ to the distal composition. Each factor is given a range and a distribution over it, so that the output becomes a random variable and its variability can be attributed to the inputs. The indices below are defined for a scalar output, so the analysis is repeated on each component of the composition; $Y = \mathcal{G}(\boldsymbol{\theta})$ denotes the component under analysis.

Two questions are asked, in this order: which factors can be left at their nominal value without changing the output, and, among the ones that are left, which matter most. Two methods are used because the two questions do not cost the same. The Morris method answers the first at a cost linear in k, and Sobol' indices answer the second on the reduced set. Both are global methods: they move the factors over their whole ranges, unlike a derivative at the nominal point, which only describes its neighbourhood and says nothing about what happens when several factors move together [9, 41].

Morris screening. The inputs are rescaled to the unit hypercube $[0,1]^k$ and sampled on a grid of L levels. The idea is to pick r starting points at random on this grid and to build a trajectory from each of them: the factors are moved one at a time, each by a step Δ along its own axis of the parameter space, until all k have been moved. A trajectory costs $k+1$ evaluations and gives one *elementary effect* per factor, the finite difference produced by its own step,

$$EE_j = \frac{\mathcal{G}(\boldsymbol{\theta} + \Delta \mathbf{e}_j) - \mathcal{G}(\boldsymbol{\theta})}{\Delta}, \qquad \mu_j^* = \frac{1}{r} \sum_{i=1}^{r} \left| EE_j^{(i)} \right|, \tag{4.1}$$

where $\mathbf{e}_j$ is the j-th canonical vector, so that only the j-th coordinate changes, and $\boldsymbol{\theta}$ is the point the trajectory has reached when the j-th factor is moved. The Morris index μ_j^*

averages the absolute values over the r trajectories [10, 30]: the absolute value is needed because effects of opposite sign at different points of the space would otherwise cancel and hide an influential factor. Since the starting points are spread over the hypercube, μ^* is a global measure built from local differences. A large value means that the factor moves the output somewhere in its range; a small one is evidence that it can be fixed. Morris only ranks the factors: it does not decompose the variance of the output, and it does not separate nonlinearity from interaction.

Sobol' indices. Variance-based indices answer the same question in a quantitative way, and they are built from one simple idea: what would happen to the uncertainty of Y if we could fix one factor? Suppose θ_i is held at some value c while all the others keep varying. The variance that is left, $\mathrm{Var}_{\boldsymbol{\theta}_{\sim i}}[Y \mid \theta_i = c]$, is a conditional variance, and it is smaller than $\mathrm{Var}[Y]$ because one source of variation has been switched off: the more it drops, the more θ_i was responsible for the spread of the output. This quantity still depends on the value c we happened to choose, so we average it over all the values θ_i can take, which gives $\mathbb{E}_{\theta_i}[\mathrm{Var}_{\boldsymbol{\theta}_{\sim i}}[Y \mid \theta_i]]$: the variance that survives, on average, when θ_i is fixed. The law of total variance splits the output variance into this term and the rest,

$$\mathbb{E}_{\theta_i}[\mathrm{Var}_{\boldsymbol{\theta}_{\sim i}}[Y \mid \theta_i]] + \mathrm{Var}_{\theta_i}[\mathbb{E}_{\boldsymbol{\theta}_{\sim i}}[Y \mid \theta_i]] = \mathrm{Var}[Y], \tag{4.2}$$

so the second term is exactly the part of the variance that fixing θ_i would remove. Dividing it by $\mathrm{Var}[Y]$ makes it a fraction between 0 and 1, and gives the first-order index [46]

$$S_i = \frac{\mathrm{Var}_{\theta_i}\left[\mathbb{E}_{\boldsymbol{\theta}_{\sim i}}[Y \mid \theta_i]\right]}{\mathrm{Var}[Y]}. \tag{4.3}$$

It is read directly as the share of the output variance carried by θ_i alone, and it needs no assumption on the shape of the model: unlike a regression coefficient, which measures how much of the variance a linear approximation of the model would explain, S_i is defined for any model, and is estimated by sampling instead of by fitting.

The S_i do not tell the whole story. They add up to one only when the model is *additive*, that is when the output can be written as a sum of terms each depending on one factor alone. Otherwise their sum is smaller than one, and what is missing is due to *interactions*: two factors interact when their joint effect on the output is not the sum of their separate effects. Conditioning on a pair of factors makes this explicit, $\mathrm{Var}[\mathbb{E}[Y \mid \theta_i, \theta_j]] = V_i + V_j + V_{ij}$, where V_i and V_j are the numerators of (4.3) and the extra term V_{ij} is the part of the response to θ_i and θ_j that neither of them explains on its own; the corresponding index is $S_{ij} = V_{ij}/\mathrm{Var}[Y]$, and it is zero when the two factors act additively. Continuing with

triples and larger groups, the whole variance is recovered,

$$\sum_i S_i + \sum_i \sum_{j>i} S_{ij} + \sum_i \sum_{j>i} \sum_{l>j} S_{ijl} + \cdots + S_{12\ldots k} = 1. \tag{4.4}$$

Every portion of the output variance is therefore attributed either to a single factor or to a group of factors acting jointly, which is what makes these indices a decomposition and not just a ranking. The problem is that this summation (4.4) has $2^k - 1$ terms; for instance, for k = 12 it has 4095 terms, too many to look at in practice.

The way around it is to group, for each factor, all the terms it appears in [19]. Fixing every factor *except* θ_i leaves the variance $\mathbb{E}_{\boldsymbol{\theta}_{\sim i}}[\mathrm{Var}_{\theta_i}[Y \mid \boldsymbol{\theta}_{\sim i}]]$, and by (4.2) applied to the group $\boldsymbol{\theta}_{\sim i}$ this is precisely what is left once the contributions that do not involve θ_i have been taken away. Normalizing gives the total-effect index

$$S_{T_i} = \frac{\mathbb{E}_{\boldsymbol{\theta}_{\sim i}}\left[\mathrm{Var}_{\theta_i}[Y \mid \boldsymbol{\theta}_{\sim i}]\right]}{\mathrm{Var}[Y]} = S_i + \sum_{j \neq i} S_{ij} + \ldots, \tag{4.5}$$

the sum of every term of (4.4) containing i, at the cost of one index per factor rather than $2^k - 1$. Three consequences are used throughout this chapter. First, $S_{T_i} \geq S_i$, and the gap between the two measures how much θ_i acts through interactions rather than on its own. Second, $\sum_i S_i \leq 1$, with equality only for an additive model, so the distance of that sum from one says how much of the variance is carried by interactions altogether. Third, $S_{T_i} = 0$ means that θ_i has no effect anywhere in its range, alone or in combination with anything else, which is the condition for fixing it.

The indices are estimated by a pick-freeze scheme, from two independent samples of size N and the matrices obtained by swapping one column at a time [40, 43]: $N(2k + 2)$ evaluations when the second order is included, against the $r(k + 1)$ of the screening, which is why Sobol' is applied only to the factors retained by Morris. Being Monte Carlo estimates, the indices carry a numerical error, reported here as bootstrap confidence intervals, and small indices are compared only in its light.

4.2. Setup of the analysis

Outputs. The output is the one defined in Section 3.1: the composition of the four biomasses of the distal lumen at excretion,

$$\mathbf{y} = \frac{\mathbf{x}_D^l(t_e)}{\|\mathbf{x}_D^l(t_e)\|_1}, \tag{4.6}$$

evaluated on a run that has met the stopping criterion (3.10), so that $\mathbf{y}$ no longer depends on the excretion time. This is the vector that (3.1) compares with the profile weights, which is why the screening is built on it; at the nominal parameter set $\mathbf{y}^\circ = (0.471, 0.253, 0.252, 0.024)$.

The four components sum to one, so they carry three degrees of freedom. The indices are computed for all of them because they show which factor moves which group, but the panels are not independent: a factor that increases one component must decrease another, and a parameter can reduce the share of its own group if the perturbation favours the others even more.

How the ranges are built. Sensitivity indices depend on the range assigned to each input [36, 42]. Since subject-specific ranges are not available for most parameters, we use one rule throughout: each degree of freedom is varied by a factor of 2 around its nominal value. The sensitivity factors are labelled by the model parameters they perturb; the auxiliary uniform variables used to generate them are only sampling coordinates. For a positive parameter θ_j we sample

$$\theta_j = \theta_j^\circ e^{u_j}, \qquad u_j \sim \mathcal{U}(-\log 2, \log 2), \tag{4.7}$$

which corresponds to a log-uniform range $[\theta_j^\circ/2,\, 2\theta_j^\circ]$. For compositional parameters, independent log-factors u_i are mapped to a nominal composition $\mathbf{p}^\circ$ by

$$p_i = \frac{p_i^\circ e^{u_i}}{\sum_{j=1}^{D} p_j^\circ e^{u_j}}, \qquad u_i \sim \mathcal{U}(-\log 2, \log 2). \tag{4.8}$$

In the sensitivity figures and tables these factors are labelled directly as p_i, for example η_i or α_i. For $D > 2$ this notation identifies the perturbation associated with one component: the physical components are still coupled by the closure and are not sampled independently. Only the differences between the underlying log-factors affect the composition, so a log-ratio spans twice the interval of the two-part case. The full construction is in Appendix A.

The only exception to the factor-2 rule is the decay rate, varied over ± 1 decade because it represents several loss processes not described explicitly by the model. The half-saturation constants K_s and K_x are excluded, being non-identifiable in the original estimation and irrelevant under saturation, while volumes, pH and conversion factors are fixed because they define the anatomical scenario. The two hydrolysis rates are treated as one factor,

having the same nominal value and model form, and τ is shared by the four groups. The resulting 30 factors are listed in Table D.1 of Appendix D.

4.3. Results and choice of the parameters

Morris screening. The screening is run on the 30 factors of Table D.1, with the only aim of separating the factors that move the composition from those that do not, so that the much more expensive variance-based analysis can be run on a small set.

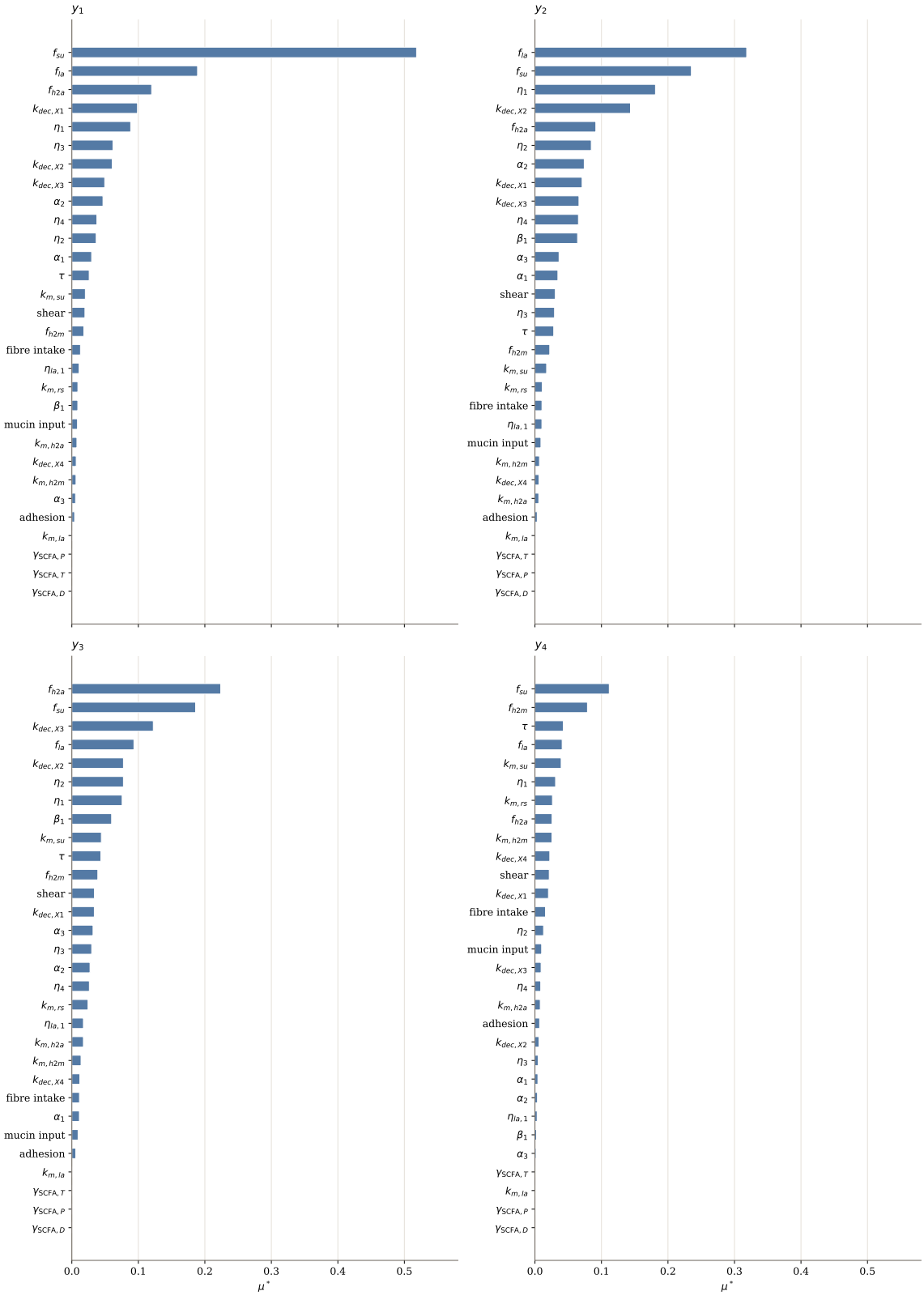


Figure 4.1: Morris index μ^* of the 30 factors, for each of the four distal fractions. Grid with $L = 4$ levels and step $\Delta = 2/3$; 60 trajectories selected out of 200 candidates by maximizing their mutual spread [10], for 1860 simulations.

The four panels are read together, keeping in mind that y_4 is on average twenty times smaller than y_1, so comparisons across panels are biased, but what matters here is the ranking. Always at the bottom, the SCFA transport coefficients and $k_{m,la}$ are negligible, with mucin secretion, adhesion, $\eta_{la,1}$, $k_{m,h2a}$ and the $\boldsymbol{\alpha}$ factors close to them. The fibre intake falls in the same region; we observed that it has a huge impact on the size of the total community, but it does not set its composition, which is what we are measuring here.

The top of the ranking is dominated by the catabolic fractions $\boldsymbol{f}$, followed by the glucose split $\boldsymbol{\eta_{su}}$ and the decay rates $\boldsymbol{k_{dec}}$. Since $1 - f$ is the growth yield, f acts directly on how much biomass a group produces from its substrate. f_{su} leads the group because it sits at the top of the trophic chain, and what it does not send to growth it sends downstream, to the groups that live on its products. The last one among the $\boldsymbol{f}$ group is f_{h2m}, which is low on the first three fractions but is kept because it is second on y_4. The glucose split follows for the same reason: it does not change how much is fermented, but where the flux goes. The decay rates act on the other side of the balance, on losses rather than gains. We keep these three blocks, twelve factors, and discard the rest.

Sobol' indices. The variance-based analysis is run on the twelve retained factors, $(f_{su}, f_{la}, f_{h2a}, f_{h2m})$, $(\eta_1, \ldots, \eta_4)$ and $(k_{dec,X_1}, \ldots, k_{dec,X_4})$, in the same ranges as the screening: a new box would produce a new ranking, and the two analyses could no longer be read as one argument. A screening also ranks factors one at a time, while the inverse problem will move a whole block at once, which is why we compute the second-order indices as well.

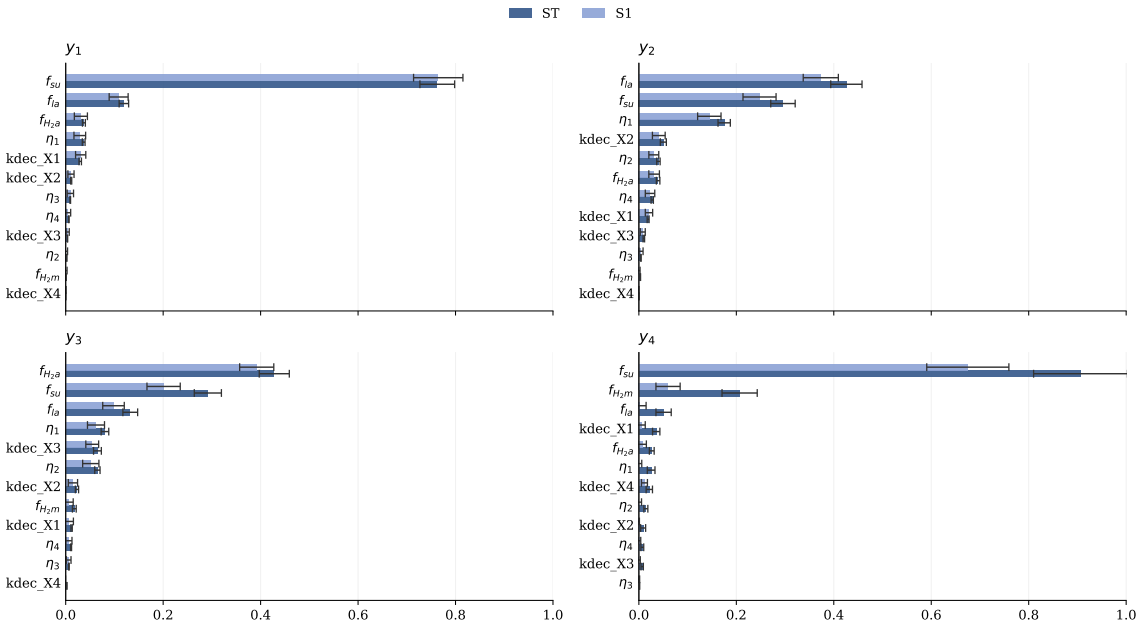


Figure 4.2: First-order (S_1, light blue) and total (S_T, dark blue) Sobol' indices of the twelve retained factors on the four distal fractions, sorted by S_T. Sobol' sequence and the estimator of [43] extended to the second order, $N(2k+2)$ runs with $N = 2048$ and $k = 12$, i.e. 53 248 simulations; error bars are bootstrap confidence intervals. The second-order indices are reported in Figure D.1.

The picture is dominated by the $\boldsymbol{f}$ block, which carries almost all the variance: summed over the block, the first-order indices give 0.91, 0.65, 0.70 and 0.74 on the four fractions. The other two blocks are marginal, and this is where the variance-based analysis corrects the screening. The decay rates never exceed $S_T = 0.07$, although their range is ten times wider than everything else. The glucose split is slightly better but still small, its largest index being η_1 on y_2 with $S_T = 0.18$ and nothing else above 0.09; being a composition, its indices are read as a block, since a common shift of the underlying log-factors changes nothing.

Interactions are weak and confined to the last fraction. On y_1, y_2 and y_3 the two bars of each factor almost coincide and the first-order indices sum to 1.00, 0.92 and 0.90: the model is additive there. On y_4 they sum to 0.75 and f_{su} opens a gap between $S_1 = 0.68$ and $S_T = 0.91$. Figure D.1 names the largest pair, (f_{su}, f_{h2m}) with $S_2 = 0.08$; the rest is spread over entries of the same order as the noise of the estimator.

Patient-specific catabolic fractions. The sensitivity analysis shows that the catabolic fractions $\boldsymbol{f}$ are the most influential block of parameters on the observed composition, but

influence alone is not enough for our purpose. A parameter may strongly affect the output when varied, yet take approximately the same value in every individual; in that case it cannot explain inter-individual differences. This is why we also require the selected parameters to be plausible carriers of *inter-individual variability*, and to act on *different* fractions, so that moving the block moves the composition along directions the outputs can tell apart.

Both requirements are met here. As described in Section 2.5, $1 - f$ is the growth yield, so each fraction controls how much of the consumed substrate a group sends to catabolism rather than to biomass. Since each microbial group aggregates different microorganisms, whose species composition changes from one subject to another, this effective metabolic allocation is expected to vary between subjects, whereas the rates k_m are properties of enzymes, the level at which inter-individual variability is least plausible. The four fractions also have distinct targets: f_{su} leads y_1, f_{la} and f_{h2a} lead y_2 and y_3, and f_{h2m} is the only factor of the design that acts on y_4 selectively, without moving the other three fractions first. The parameters estimated in the inverse problem of Chapter 5 are then:

$$\boldsymbol{\theta} = \big(f_{su},\ f_{la},\ f_{h2a},\ f_{h2m}\big). \tag{4.9}$$

This choice has two limitations. First, the block contains four parameters while the observed composition has only three degrees of freedom, so the inverse problem is underdetermined by one direction and requires regularization; the weakest direction is the one carried by f_{h2m}, which acts on y_4 only, the smallest fraction and the one where the interactions sit. Second, the Sobol' indices are global averages over the ranges explored here: they justify the choice of the block, but do not certify that all four parameters are identifiable at the solution of a given patient. The inversion of Chapter 5 does not resolve this point, it works around it: the prior supplies the direction the data leave undetermined, and the results are read through the reconstructed compositions rather than through the posterior value of each single parameter.

5 | Inverse problem

5.1. Introduction

The model of the previous chapters solves a *forward problem*: given a set of parameters, it returns a distal microbial composition. We take into account now the *inverse problem* , in which the parameters are estimated from observed data [21]. Writing θ for the unknown parameters, $\mathcal{G}(\theta)$ for the model output and z for the observation,

$$z = \mathcal{G}(\theta) + \varepsilon, \tag{5.1}$$

where ε collects measurement errors and model inaccuracies. Since the data are compositions, the sum is meant in the log-ratio coordinates used in the likelihood below, so that the error acts on the ratios between the components and not on the raw fractions.

Non-uniqueness. In our case the solution is not unique. We estimate the four parameters selected in 4.3 from a composition of four parts, which has only three independent degrees of freedom; therefore, we expect that different parameter vectors give the same output. The data alone cannot choose between them, and some information from outside the data is needed. The classical way to add it is to solve a least-squares problem with a penalty term,

$$\min_{\theta} \ \tfrac{1}{2} \left\| z - \mathcal{G}(\theta) \right\|^2 + \tfrac{\delta}{2} \left\| \theta - \theta_0 \right\|^2, \tag{5.2}$$

where θ_0 is a guess for the parameters and $\delta > 0$ weights how much we trust it. This is Tikhonov regularization: the penalty removes the ambiguity, but the answer is a single vector and it depends on δ and on θ_0, which are chosen by hand. The Bayesian approach used here follows the same idea, but the extra information and the answer are written as a probability distribution.

Bayesian inference. In the Bayesian framework the unknown parameters are treated as random variables [21]. What is known before the measurement is described by the *prior* $\pi_{\text{prior}}(\theta)$, the data enter through the *likelihood* $\pi(z \mid \theta)$, which is read as a function

of θ with the observation z fixed, and Bayes' theorem combines the two into the *posterior*

$$\pi(\theta \mid z) = \frac{\pi(z \mid \theta)\, \pi_{\text{prior}}(\theta)}{\displaystyle\int \pi(z \mid \theta)\, \pi_{\text{prior}}(\theta)\, d\theta}, \qquad \text{posterior} \propto \text{likelihood} \times \text{prior}. \tag{5.3}$$

The denominator does not depend on θ: it is the probability of the data under the model, and it only normalizes the result. The answer is a distribution and not a single value, which suits a problem whose solution is not unique: where the data say nothing, the posterior stays wide. It can be shown that, with a Gaussian likelihood and a Gaussian prior, the parameter that maximizes the posterior is the solution of the penalized problem (5.2), the prior playing the role of the penalty term. The prior is not an arbitrary addition: it is the same information that regularization introduces, written in a form that also says how strong it is.

MCMC. The main hurdle when it comes to evaluating the posterior is to determine the integral in the denominator of (5.3). That integral runs over the whole parameter space; it can be computed in closed form only in special cases, and numerical quadrature is practical only in very low dimension, because it needs a grid over a region where the posterior is concentrated, which is part of what we are looking for. For this reason, instead of computing the posterior as a function, a set of parameter vectors distributed as the posterior is generated, and every quantity of interest is approximated by an average over these samples. Drawing independent samples is not possible here, since the posterior is known only through its shape. The idea of Markov Chain Monte Carlo (MCMC) is to give up independence and to build instead a sequence $\theta^{(1)}, \theta^{(2)}, \dots$ in which each state is drawn from a rule that depends only on the current one [16]. Such a sequence is a Markov chain, and it is a convenient object because its long-run behaviour is controlled by a well developed theory [14]. Write $p(\theta, \phi)$ for the density of moving from θ to ϕ in one step. A distribution is *invariant* for the chain if, once the state is distributed in that way, it stays so at the next step. It can be shown that a chain that can reach any region of the parameter space, and that does not return to the same regions with a fixed period, has one invariant distribution only, and that the distribution of the states converges to it whatever the starting point. Building a sampler means building a chain whose invariant distribution is the posterior. A sufficient condition for this is the detailed balance condition

$$\pi(\theta \mid z)\, p(\theta, \phi) = \pi(\phi \mid z)\, p(\phi, \theta), \tag{5.4}$$

which asks that the chain moves between two states equally often in the two directions. It is easier to impose than invariance itself: rather than choosing p directly, a move

is proposed from any convenient distribution and is then accepted only with a certain probability, which is chosen so that (5.4) holds. The Metropolis algorithm is the simplest choice of this kind.

The Metropolis algorithm. A candidate is proposed by adding a random step ξ to the current state $\theta^{(n)}$, and is accepted with probability α:

$$\theta^* = \theta^{(n)} + \xi, \qquad \alpha = \min\left(1,\ \frac{\pi(\theta^* \mid z)}{\pi(\theta^{(n)} \mid z)}\right), \tag{5.5}$$

otherwise the current state is kept [29]. The step ξ is drawn from a distribution that is symmetric around zero, usually a Gaussian with a fixed covariance, which is what makes the simple ratio above the right acceptance probability. Moves towards higher posterior values are always accepted and moves towards lower values are sometimes accepted, so the chain spends most of its time where the posterior is large. It can be shown that this rule satisfies (5.4), so the posterior is the invariant distribution of the chain. This method is usable in practice thanks to the fact that only the ratio of two posterior values is needed, so that the denominator in (5.3) cancels out. Although the states are not independent, the law of large numbers and the central limit theorem still hold for averages along the chain, so these averages estimate posterior quantities and their error can be assessed. The price of the dependence is that consecutive states carry overlapping information, which is why the number of *effective* samples, rather than the number of draws, is reported together with a convergence diagnostic comparing chains started from different points [48].

The sampler used here. In a plain Metropolis scheme the distribution of the step ξ has to be tuned by hand, and a poor choice explores a correlated posterior very slowly. DEMetropolisZ [47] keeps the acceptance rule (5.5) and only changes how the step is built, using states visited earlier by the chain,

$$\theta^* = \theta^{(n)} + \underbrace{\lambda\left(\theta^{(r_1)} - \theta^{(r_2)}\right) + \xi}_{\text{step}}, \tag{5.6}$$

so that the size and the direction of the moves follow the shape of the posterior without any tuning. The two past states are picked in the same way, so the step is again symmetric around zero and (5.5) applies unchanged. Strictly speaking the sequence is no longer a Markov chain, since each step uses the whole history and not only the current state. This is the same situation as in adaptive MCMC, and the reason why the method is still valid is that the Markov property is a means and not the goal: what the estimates need is only

that the averages along the chain converge to posterior averages. It can be shown that this holds when the dependence on the past becomes weaker and weaker as the history grows [38], which is the case here, since each new state changes an ever larger archive less and less. An alternative is NUTS [18], which uses the gradient of the log-posterior to propose distant states, possible here because the surrogate of Section 5.3 is differentiable; the two are compared in Appendix E.5. The implementation and the sampler settings are given in Appendix E.4.

5.2. Setup of the inversion

Following the sensitivity analysis of Chapter 4, we vary only the four catabolic fractions

$$\theta = (f_{\mathrm{su}},\ f_{\mathrm{la}},\ f_{\mathrm{h2a}},\ f_{\mathrm{h2m}}) \in [0.01, 0.99]^4,$$

and keep all the other parameters at their nominal values. The observed data are the four functional profile weights, normalized as a composition, while the model output is the distal biomass composition defined in Section 3.1.

Parameter range. This box is much wider than the one used in Chapter **??**. In that analysis, however, the range had to be chosen consistently across parameters, because Sobol' indices depend on the range assigned to each factor. Ranking thirty heterogeneous factors therefore requires the same relative perturbation for all of them; for the maximum specific rates or the intakes there is no maximal range anyway, since they are not bounded from above, and using one would have measured the width of the boxes instead of the influence of the factors. Here the parameters are not being ranked against each other. Since $1 - f$ is the growth yield, a catabolic fraction lives in $0 < f < 1$ by construction, so $[0.01, 0.99]$ is not a choice but the domain of the parameter, the margins only avoiding the degenerate limits of zero growth and zero catabolism.

Likelihood. Model and data are compared with the Aitchison distance. With the surrogate $\widehat{\mathcal{G}}$ of Section 5.3, the likelihood is

$$\log \pi(z \mid \theta) \propto -\frac{1}{2\sigma^2} d_a^2\left(\widehat{\mathcal{G}}(\theta), z\right), \qquad \sigma = 0.2, \tag{5.7}$$

where z is one observed composition. The value of σ is fixed and is used as a tolerance for the difference between model and data.

Priors. The priors are independent truncated Gaussians,

$$\theta \sim \mathcal{N}_T\left(\mu,\, 0.25^2 I;\, [0.01, 0.99]^4\right), \qquad \mu = (0.90,\, 0.80,\, 0.57,\, 0.38)^T, \tag{5.8}$$

centred on the reference values of Muñoz-Tamayo et al. [31].

Weights below 10^{-6} are treated as absent profiles and replaced by the threshold, as described in Section 1.2. When a profile is absent, the parameter mainly acting on that component should not be interpreted as directly constrained by a measured abundance.

Along the directions where the output does not change, the likelihood is flat and the posterior follows the prior: this is what makes the solution unique. The aim is not to recover a unique biological parameter vector, but to find model configurations that reproduce the observed data.

5.3. Surrogate model

The inverse problem needs many evaluations of the model. One simulation of the full model takes tens of seconds, so repeating it inside MCMC for all the samples would be too expensive. For this reason we use a neural surrogate of the forward map. The surrogate receives the four catabolic fractions and returns the distal microbial composition,

$$\widehat{\mathcal{G}}:\ [0.01, 0.99]^4 \longrightarrow \mathcal{S}^3, \qquad (f_{\mathrm{su}}, f_{\mathrm{la}}, f_{\mathrm{h2a}}, f_{\mathrm{h2m}}) \longmapsto \mathbf{y} = (y_1, y_2, y_3, y_4).$$

The output is a composition, so the error is measured with the Aitchison distance introduced in Appendix A.

Training data. The training design contains $2^{15} = 32768$ points in the box $[0.01, 0.99]^4$, generated with a scrambled Sobol' sequence [33, 45]. This gives a more uniform coverage of the parameter box than independent random sampling. Each point is evaluated with the full model using the stopping rule of Section 3.4. Details on the design, network and training procedure are given in Appendix E.1 and Appendix E.2.

Network. The surrogate is a feed-forward neural network, a multilayer perceptron with four hidden layers of 128 units and tanh activations, about $5{\cdot}10^4$ parameters, followed by a softmax layer that places the prediction in $\mathcal{S}^3$ by construction. It is trained by minimising the squared Aitchison distance between predicted and simulated compositions. The map is deterministic and smooth, which is why a plain perceptron of this size is enough; the schematic, the training settings and the implementation are in Appendix E.2.

Accuracy. The surrogate is accurate on the test set. The median Aitchison error is 0.012 and the 95th percentile is 0.060. The coefficient of determination R^2 in clr coordinates is above 0.998 for all four components. The surrogate error is small compared with the variability of the model outputs and of the observed data. The agreement between predicted and simulated compositions is shown in Figure E.2 of Appendix E.2.

The surrogate is used only as a computational approximation of the mechanistic model. For this reason, a final check with the full model is performed after the dataset reconstruction in Section 5.5.

5.4. Reconstruction of a representative sample

We first test the complete procedure on one representative sample. Instead of using the mean composition, which does not correspond to a real observation, we select the medoid of the 3613 cleaned samples. The medoid is the observed composition with the smallest average Aitchison distance from all the others.

Sampling. For this case we use six long chains started from different points of the parameter box; the settings are in Table E.2. All four parameters reach $\hat{R} \leq 1.002$ [48], with more than 7800 effective samples and no convergence warning.

Reconstruction. The main quantity of interest is the reconstructed composition. All four observed fractions fall inside the 95 % posterior predictive intervals in Figure 5.1. The predictive median differs from the observation by less than 6 % on every component. The median Aitchison distance between a predictive *draw* and the observation is 0.30: this is the width of the predictive distribution, not the distance between the predictive median and the data. For comparison, the observed samples lie at an average Aitchison distance of 3.24 from their compositional centre.

This means that the sample is reproduced by the model. The parameter values are less important than the reconstructed composition: several combinations of the four fractions give almost the same fit. This is expected from the dimension of the inverse problem and means that the result should not be read as a unique estimate of the true catabolic fractions of the sample.

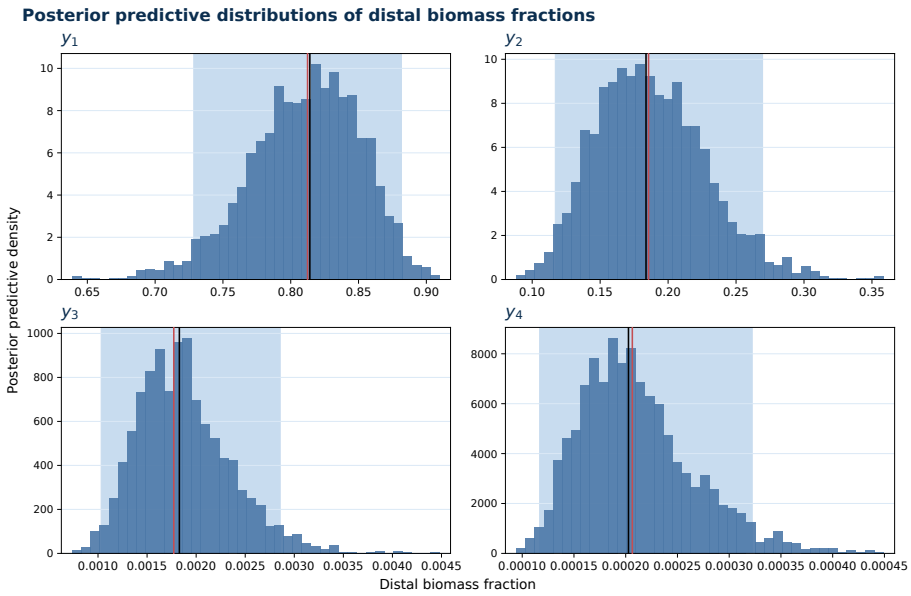


Figure 5.1: Posterior predictive distribution for the representative sample. The shaded region is the 95 % interval, the black line is the predictive median and the red line is the observed composition.

5.5. Reconstruction across the dataset

The same procedure is then applied independently to all 3613 cleaned compositions, with the shorter runs of Table E.2, chosen to make the reconstruction of the full dataset possible in a few hours. They are used to find compatible model configurations, not to make a detailed uncertainty analysis of each individual parameter. The whole calculation takes 2 h 46 min on 64 cores; see Appendix E.6.

Predictive spread. The agreement with the data is similar over the whole dataset. The Aitchison distance between a posterior predictive draw and the observation has mean 0.333, median 0.331 and maximum 0.563. These numbers measure the width of the predictive distribution rather than the distance between model and data: d_a is a Euclidean norm on the three-dimensional clr space, so whenever the model can pass close to the observation the likelihood (5.7) concentrates this distance around $\sigma\sqrt{3} \approx 0.35$, which is the scale observed here. For comparison, the observed samples lie at an average Aitchison distance of 3.24 from their compositional centre, so the predictive distribution is an order of magnitude narrower than the variability present in the data.

This result does not prove that the model gives the true internal state of each individual. It shows that the observed functional compositions can be closely approximated, over the whole cleaned dataset, by changing only the four catabolic fractions selected in Chapter 4.

Check with the full model. The distances above are obtained with the neural surrogate and average over the posterior. The misfit itself is measured at a single parameter vector: for one hundred samples selected at random we take the posterior median of the four fractions and simulate them again with the full mechanistic model. The median Aitchison distance between the full-model output and the observation is 0.142, with most values between about 0.05 and 0.25 and a small tail up to 0.43. This is the distance between model and data, not the predictive spread of the previous paragraph; the protocol is in Appendix E.3.

	samples	mean	median	max
Predictive spread (surrogate)	3613	0.333	0.331	0.563
Misfit at posterior median (full model)	100	0.166	0.142	0.429

Table 5.1: Aitchison distances. The first row is the distance between a posterior predictive draw and the observation, over the complete cleaned dataset: it measures the width of the predictive distribution, whose scale is set by the tolerance σ of (5.7). The second row is the distance between model and data at the posterior median of the parameters, for one hundred randomly selected samples re-simulated with the full mechanistic model. The two rows are different quantities and should not be read as a comparison.

The parameter values found through the surrogate remain very close to the observed compositions when they are evaluated with the original model.

What the inversion gives. The final result is a sample-specific model reconstruction. Starting from one observed functional composition, the procedure finds a configuration of the mechanistic model that reproduces it. The configuration is not unique, because the data contain fewer independent quantities than the number of parameters. For this reason, the cohort results are interpreted through the reconstructed compositions and not through the posterior value of each single parameter.

This is the main role of the inverse problem in this work: it creates a bridge from real microbiome data to the mechanistic model. Once a sample has been placed inside the model, the model can be simulated or perturbed to study quantities that are not directly measured in the stool sample.

6 | Conclusions and future developments

This thesis builds a workflow that connects real microbiome data with a mechanistic model of the colon. The data give four functional profile weights for each faecal sample. The model gives four distal microbial biomasses. The main modelling assumption of the work is that their relative compositions can be compared, as written in Eq. (3.1). This assumption creates the link between data and model, but it is not a direct measurement of biomass.

The workflow has three main steps. First, the original microbial groups of the Mu noz-Tamayo model [31] are adapted to match the four functional profiles. Second, global sensitivity analysis is used to select the parameters with the strongest effect on the distal composition. The four catabolic fractions are selected. Third, an inverse problem is solved with a neural surrogate of the model, allowing the procedure to be repeated on the full dataset.

The main result is the reconstruction of the observed compositions. Across the 3613 cleaned samples the posterior predictive distribution stays within a median Aitchison distance of 0.331 from the observation, an order of magnitude below the spread between subjects (3.24). Evaluated at the posterior median of the parameters, and with the full mechanistic model in place of the surrogate, the median distance between model and data is 0.142: the agreement is therefore not an artifact of the surrogate.

The result must be interpreted with care. A four-part composition has only three independent degrees of freedom, while four parameters are varied; one observed composition cannot identify a unique parameter vector. The fitted fractions should not be read as the true biological values of one individual. The same problem is stronger when a functional profile is absent and has been replaced by the numerical threshold used in the data cleaning.

Thanks to the contribution of the inverse problem one observed sample can be associated with a model configuration that reproduces its functional composition. This gives a way

to place real microbiome data inside the mechanistic model. The fitted model can then be used to simulate quantities that are not directly measured, or to study its response to a perturbation.

Richer physical models. A natural continuation is to apply the same workflow to richer models. Later work in the group includes spatial models and more detailed descriptions of the interaction between microbiota and host [15, 34]. These models have many more parameters and are more expensive to simulate. In that case, sensitivity analysis and surrogate models become even more useful.

Richer patient data. The data used here mainly contain the relative functional composition. Other measurements would give different information. Faecal SCFA concentrations and total biomass would constrain the absolute metabolic state. Repeated samples would give information on time variability. Diet would be especially useful when the fitted model is used to study the response to a dietary change.

Large projects such as Le French Gut [11] may provide this type of richer information. Another possible direction is a hierarchical Bayesian model, where information is shared between samples or clinical groups instead of solving every sample completely independently.

Mechanistic models and machine learning. The four functional profiles are a strong compression of the original metagenomic data. This makes them easy to connect with a small mechanistic model, but much of the information in the DNA is lost. Methods such as MetagenBERT [39] use a much richer representation and can find signals that are not visible in four dimensions.

The two approaches answer different questions. Machine learning can be useful to detect a signal in large biological datasets. A mechanistic model is useful when we want to connect that signal with processes such as substrate consumption, microbial growth and metabolite production. A useful future direction is therefore to combine the two approaches.

Closing remark. A faecal sample is only the final output of a process that happens inside the colon and is difficult to observe directly. This thesis does not recover that process uniquely. It shows that real functional compositions can be connected to a mechanistic model and reproduced by it. With richer data and richer models, the same workflow could give a more detailed description of the differences between individuals.

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A | Compositional data

Several parts of this work describe proportions rather than absolute amounts: the NMF profiles of Chapter 1 and the microbial fractions produced by the model are examples. They are *compositional data* and require a specific geometry [2, 3, 35].

A.1. The simplex

A composition with D parts is a vector of positive numbers that sums to one. Its sample space is the *simplex*

$$\mathcal{S}^{D-1} = \left\{ x = (x_1, \ldots, x_D) \in \mathbb{R}^D_{>0} : \sum_{i=1}^{D} x_i = 1 \right\}, \tag{A.1}$$

and any positive vector is mapped to it by the *closure* operator

$$\mathcal{C}(x) = \left(\frac{x_1}{\sum_j x_j}, \ldots, \frac{x_D}{\sum_j x_j} \right). \tag{A.2}$$

Closure removes the total and keeps only the relative information: $(2, 1, 1)$ and $(20, 10, 10)$ give the same composition. Because of the constant-sum constraint the simplex has $D-1$ degrees of freedom and its components cannot vary independently, so compositional data should not be treated as ordinary vectors in $\mathbb{R}^D$ [3, 35].

A.2. Aitchison geometry

Aitchison geometry gives the simplex the structure of a vector space [3, 35], with the *perturbation* playing the role of addition,

$$x \oplus y = \mathcal{C}(x_1 y_1, \ldots, x_D y_D), \tag{A.3}$$

and the inner product and distance

$$\langle x, y\rangle_a = \frac{1}{2D}\sum_{i=1}^{D}\sum_{j=1}^{D}\log\frac{x_i}{x_j}\log\frac{y_i}{y_j}, \qquad d_a(x,y) = \|x \ominus y\|_a, \tag{A.4}$$

where $\ominus$ denotes perturbation by the opposite composition. This distance depends only on ratios between components and is therefore consistent with the relative nature of compositional data.

A.3. The centred log-ratio transformation

The same geometry can be written in ordinary Euclidean coordinates using log-ratios [2]. With $g(x) = \left(\prod_{i=1}^{D} x_i\right)^{1/D}$ the geometric mean of the components, the *centred log-ratio* (clr) transformation is

$$\mathrm{clr}(x) = \left(\log\frac{x_1}{g(x)}, \ldots, \log\frac{x_D}{g(x)}\right), \qquad \sum_{i=1}^{D}\mathrm{clr}_i(x) = 0. \tag{A.5}$$

It keeps one coordinate for each component, which makes it easy to interpret, although the coordinates are not independent because they sum to zero. Aitchison distance becomes the usual Euclidean distance in clr space,

$$d_a(x,y) = \|\mathrm{clr}(x) - \mathrm{clr}(y)\|_2, \tag{A.6}$$

which is the form used throughout this work. For clr coordinates c, with $\sum_i c_i = 0$, the inverse mapping is obtained by exponentiating and closing,

$$\mathcal{C}\left(e^{c_1}, \ldots, e^{c_D}\right) = \mathrm{softmax}(c), \tag{A.7}$$

and adding the same constant to every component of c does not change the resulting composition.

A.4. Perturbing a composition

The sensitivity analysis of Chapter 4 requires independent input factors, while the components of a composition cannot be sampled independently. We therefore sample independent multiplicative factors and use the perturbation to map them back to the simplex.

Let $p^\circ \in \mathcal{S}^{D-1}$ be a nominal composition and let $u_i = e^{z_i}$ be a positive vector of factors.

Perturbing p° by $\mathcal{C}(u)$ gives, component-wise,

$$p_i = \frac{p_i^\circ u_i}{\sum_j p_j^\circ u_j} = \frac{p_i^\circ e^{z_i}}{\sum_{j=1}^{D} p_j^\circ e^{z_j}}, \tag{A.8}$$

which is always positive and sums to one. The effect of the perturbation is clearer on the ratios,

$$\frac{p_i}{p_j} = \frac{u_i}{u_j}\,\frac{p_i^\circ}{p_j^\circ}, \tag{A.9}$$

so the factors act multiplicatively on the relative information carried by the composition. The z_i can be sampled independently in an ordinary box, while (A.8) enforces the compositional constraint. This is the mapping used in Chapter 4.

Two-part compositions. A two-part composition $(p, 1-p)$ has only one degree of freedom, so the second factor is kept fixed and we sample $z \sim \mathcal{U}(-\log 2, \log 2)$ with $u = (e^z, 1)$. Equation (A.8) becomes

$$p = \frac{p^\circ e^z}{p^\circ e^z + 1 - p^\circ}, \qquad \text{equivalently} \qquad \log\frac{p}{1-p} = \log\frac{p^\circ}{1-p^\circ} + z, \tag{A.10}$$

so the uniform variable acts directly on the log-ratio between the two parts, and the fraction p itself is sampled neither uniformly nor log-uniformly. The two limits, obtained with $e^z = 1/2$ and $e^z = 2$, are

$$p_{\min} = \frac{p^\circ}{2 - p^\circ}, \qquad p_{\max} = \frac{2p^\circ}{1 + p^\circ}. \tag{A.11}$$

Compositions with more than two parts. For $D > 2$ we use the same rule, with one factor $z_i \sim \mathcal{U}(-\log 2, \log 2)$ sampled independently for each component, and recover the composition through (A.8).

Although D factors are sampled, the composition still has only $D-1$ degrees of freedom: adding the same constant to every z_i leaves (A.8) unchanged, so a common shift is invisible to the model and the indices of such a block should be read together rather than one factor at a time. We keep all D factors because this avoids choosing one component as a fixed reference and treats all components in the same way. The price is that a log-ratio $\log(p_i/p_j)$ now varies over $[-2\log 2,\ 2\log 2]$, twice the interval of the two-part case: multi-part compositions are given a slightly wider box than the other factors. The independent box is therefore defined in the variables z_i, while the components p_i passed to the model

are dependent and automatically satisfy the simplex constraint.

A.5. Centre and total variance

The mean of a set of compositions $x^{(1)}, \dots, x^{(N)}$ is not the arithmetic mean of the parts. The *centre* is the closed geometric mean,

$$\operatorname{cen}(x) = \mathcal{C}\Big(\big(\textstyle\prod_i x_1^{(i)}\big)^{1/N}, \dots, \big(\textstyle\prod_i x_D^{(i)}\big)^{1/N} \Big),$$

which is the arithmetic mean in clr coordinates mapped back to the simplex. The dispersion of the sample is measured by the *total variance*

$$\operatorname{totvar}(x) = \frac{1}{N} \sum_{i=1}^{N} d_a\big(x^{(i)}, \operatorname{cen}(x)\big)^2,$$

the mean squared Aitchison distance to the centre. It gives the natural scale against which an error expressed in Aitchison distance should be compared.

A.6. Zeros

All log-ratio transformations require strictly positive components, so zeros must be treated before applying the clr transformation or the Aitchison distance. Two cases are usually distinguished [28]: a *rounded zero* is a small positive value recorded as zero because it is below a detection limit, and can be replaced by a small positive value, while an *essential zero* represents the real absence of a component and has a different meaning. The treatment used for the microbiome profiles in this work is described in Section 1.2.

B | Model Nomenclature, Parameters and Supporting Material

B.1. Purpose of this appendix

This appendix collects the technical material associated with the reference mathematical model introduced in Chapter 2. The main text focuses on the structure of the formulation and on the interpretation of the parameters, whereas the present appendix gathers the detailed notation, the baseline parameter values, the list of chemical reactions, the kinetic expressions and the derivation of the yield factors, that is everything needed to reconstruct the model in full.

The symbols, indices and numerical values reported here follow strictly the nomenclature of [31], so that every equation of Chapter 2 can be compared one-to-one with the original formulation.

B.2. Nomenclature

In each of the six compartments the state vector is

$$\zeta = \left(\mathbf{s}^{\mathrm{T}},\, z,\, \mathbf{x}^{\mathrm{T}},\, \mathbf{s}_g^{\mathrm{T}}\right)^{\mathrm{T}},$$

and ξ denotes the vector of concentrations in the liquid phase. The superscripts l and m refer to the lumen and the mucus microhabitats respectively.

State variables

Symbol	Meaning
s_i	concentration of soluble component i in liquid phase, expressed in M (mol/l)
$s_{g,i}$	concentration of soluble component i in gas phase (M)
x_i	concentration of bacterial group i (M)
z	concentration of polysaccharides (g/l)
ξ	vector of concentrations in the liquid phase
ζ	full state vector

Table B.1: State variables.

The soluble components are, in the order used by the Petersen matrix,

$$\mathbf{s} = \left(s_{\mathrm{su}},\, s_{\mathrm{la}},\, s_{\mathrm{H_2}},\, s_{\mathrm{ac}},\, s_{\mathrm{pro}},\, s_{\mathrm{bu}},\, s_{\mathrm{CH_4}},\, s_{\mathrm{CO_2}},\, s_{\mathrm{H_2O}}\right)^{\mathrm{T}},$$

the microbial functional groups are

$$\mathbf{x} = \left(x_{\mathrm{su}},\, x_{\mathrm{la}},\, x_{\mathrm{H_2a}},\, x_{\mathrm{H_2m}}\right)^{\mathrm{T}},$$

and the gas-phase components are

$$\mathbf{s}_g = \left(s_{g,\mathrm{H_2}},\, s_{g,\mathrm{CH_4}},\, s_{g,\mathrm{CO_2}}\right)^{\mathrm{T}}.$$

Parameters

Symbol	Meaning
a_i	adherence coefficient of bacteria in lumen (d^{-1})
b_i	shear loss coefficient of bacteria in mucus (d^{-1})
k_d	decay constant of bacteria (d^{-1})
$K_{H,i}$	Henry's law coefficient of component i (M/bar)
$k_{\mathrm{hyd},i}$	maximum specific hydrolysis rate (d^{-1})
$k_L a$	liquid–gas transfer coefficient multiplied by the specific transfer area (d^{-1})
$k_{m,i}$	maximum specific rate of substrate consumption (mol s_j d^{-1}/mol x_j).
	For hydrolysis, it is given in g s_j d^{-1}/mol x_j
$K_{s,i}$	half-saturation constant for microbial growth (Monod's constant) (M)
$K_{x,i}$	half-saturation coefficient for ratio s/x (M s_j / M x_j)
pH_{L}	lower limit of pH inhibition
pH_{U}	upper limit of pH inhibition
ρ_i	density of component i (g/l)
w_i	molecular weight of component i (g/mol)
$Y_{i,j}$	yield of component i in process j (mol s_i/mol s_j).
	For hydrolysis, it is given in mol s_i/g s_j
Y_j	biomass yield factor for bacteria utilizing substrate j (mol x_j/mol s_j)
γ_i	transport coefficient of component i (d^{-1}). For sugars, it is given in l/d
τ_i	additional residence time of bacteria in lumen (d)

Table B.2: Parameters. Note that, as in [31], the symbol ρ is used both for the densities ρ_i and for the kinetic rates $\rho_j(\xi)$; the two are distinguished by the index, i running over components and j over processes.

Functions

Symbol	Meaning
ρ_j	kinetic rate of process j
Q_i	liquid–gas transfer rate of component i
q_n	net flow rate in the distal lumen (l/d)
q_{out}	effluent flow rate of the lumen compartments (l/d)
q^m_{out}	flow rate of mucins to lumen (l/d)

Table B.3: Functions.

Subscripts and superscripts

General		**For soluble components**	
g	gas phase	ac	acetate
in	influent	bu	butyrate
l	lumen	la	lactate
lgt	liquid–gas transfer	pro	propionate
m	mucus	su	sugar (glucose)
out	effluent		
n	net		
For particulate components			
H_2a	hydrogen-utilizing bacteria (acetogenic)	la	lactate-utilizing bacteria
H_2m	hydrogen-utilizing bacteria (methanogenic)	su	glucose-utilizing bacteria

Table B.4: Subscripts and superscripts.

Physiological parameters

Symbol	**Meaning**
P_{atm}	atmospheric pressure (bar)
p_{H_2O}	water pressure (bar)
R	ideal gas constant (bar/(M K))
q	flow rate (l/d)
T	temperature (K)
v_m	volume in gel phase of mucosal sections (l)
v_g	volume in gas phase of mucosal sections (l)
V_l	volume in liquid phase of luminal sections (l)
V_g	volume in gas phase of luminal sections (l)
Γ	endogenous production of mucins (g/d)

Table B.5: Physiological parameters.

B.3. Numerical values of the parameters

Approximate dimensions of the human colon

Anatomical region	Length (cm)	Volume of lumen microhabitat (l)	Volume of mucus[a] microhabitat (l)
Cecum	6		
Ascending colon	15		
Proximal colon	21	0.41	0.017
Transverse colon	50	0.98	0.042
Descending colon	25		
Sigmoid colon	40		
Rectum	18		
Distal colon	83	1.63	0.070
Total	154	3.02	0.129

Table B.6: Approximate dimensions of the human colon. [a]The volume of the mucus $(v_m + v_g)$ is calculated assuming cylindric shape as $v_m + v_g = 0.25\pi\left(\phi^2 - (\phi - 2e_m)^2\right)L$. The thickness of the mucus e_m is taken to be equal approximately to 0.0830 cm; ϕ is the diameter of the colon (5 cm), and L is the length of each section. The volume of the lumen is calculated as $0.25\pi\phi^2 L - (v_m + v_g)$.

Physiological parameters

Physiological parameter	Value	Comments
P_{atm}	1.013 bar	
pH	5.5^P, 6.2^T, 6.8^D	Superscripts P, T, D denote proximal colon, transverse colon and distal colon
p_{H_2O}	0.08274 bar	
R	0.08314 bar/(M K)	
T	310.15 K	
q_{in}	1.5 l/d	Input flow in the proximal lumen
V_g, v_g	—	It is assumed that the gas phase occupies 10% of the total volume in each microhabitat (but for the lumen in distal colon, where the volume in both phases changes over time)
Γ	5 g/d	Each of the three sections is assumed to produce $\Gamma/3$

Table B.7: Physiological parameters.

Kinetic parameters and yield factors

Parameter	Value	Parameter	Value
$k_{\mathrm{hyd},z}$	1.20×10^3	$Y_{\mathrm{pro,la}}$	0.267
$K_{x,z}$	29.99	$Y_{\mathrm{bu,la}}$	0.200
$Y_{\mathrm{su},z}$	0.005	$Y_{\mathrm{CO_2,la}}$	0.533
$k_{m,\mathrm{su}}$	7.92	$Y_{\mathrm{H_2O,la}}$	0.493
$K_{s,\mathrm{su}}$	0.0026	$k_{m,\mathrm{H_2a}}$	108.837
Y_{su}	0.120	$K_{s,\mathrm{H_2a}}$	0.0017
$Y_{\mathrm{la,su}}$	0.499	$Y_{\mathrm{H_2a}}$	0.043
$Y_{\mathrm{H_2,su}}$	1.440	$Y_{\mathrm{ac,H_2a}}$	0.143
$Y_{\mathrm{ac,su}}$	0.567	$Y_{\mathrm{CO_2,H_2a}}$	-0.5
$Y_{\mathrm{pro,su}}$	0.240	$Y_{\mathrm{H_2O,H_2a}}$	0.629
$Y_{\mathrm{bu,su}}$	0.270	$k_{m,\mathrm{H_2m}}$	22.581
$Y_{\mathrm{CO_2,su}}$	1.100	$K_{s,\mathrm{H_2m}}$	1.563×10^{-6}
$Y_{\mathrm{H_2O,su}}$	1.440	$Y_{\mathrm{H_2m}}$	0.062
$k_{m,\mathrm{la}}$	10^3	$Y_{\mathrm{CH_4,H_2m}}$	0.095
$K_{s,\mathrm{la}}$	6.626×10^{-3}	$Y_{\mathrm{CO_2,H_2m}}$	-0.450
Y_{la}	0.120	$Y_{\mathrm{H_2O,H_2m}}$	0.686
$Y_{\mathrm{H_2,la}}$	0.400	k_d	0.01
$Y_{\mathrm{ac,la}}$	0.133	pH_{L}, pH_{U}	5.8, 6.7

Table B.8: Kinetic parameters and yield factors. Kinetic parameters are the same in all compartments.

Parameters associated to mass-transfer phenomena

Parameter	Value
k_La	200 d^{-1}
$K_{H,\mathrm{H_2}}$	7.29×10^{-4} M/bar
$K_{H,\mathrm{CO_2}}$	0.0255 M/bar
$K_{H,\mathrm{CH_4}}$	0.0011 M/bar

Table B.9: Liquid–gas transfer parameters.

Parameter	Proximal colon	Transverse colon	Distal colon
γ^l_{su}	1.60 l/d	3.80 l/d	6.30 l/d
γ^l_{la}	0.88 d^{-1}	0.43 d^{-1}	2.03 d^{-1}
γ^l_{ac}	1.32 d^{-1}	0.64 d^{-1}	3.05 d^{-1}
γ^l_{pro}	1.07 d^{-1}	0.62 d^{-1}	2.47 d^{-1}
γ^l_{bu}	0.90 d^{-1}	0.57 d^{-1}	2.49 d^{-1}
$\gamma^l_{\text{H}_2\text{O}}$	1.60 d^{-1}	0.77 d^{-1}	3.66 d^{-1}
γ^m_{su}	1.60 l/d	3.80 l/d	6.30 l/d
γ^m_{la}	12.60 d^{-1}	12.60 d^{-1}	12.60 d^{-1}
γ^m_{ac}	18.90 d^{-1}	18.90 d^{-1}	18.90 d^{-1}
γ^m_{pro}	15.32 d^{-1}	15.32 d^{-1}	15.32 d^{-1}
γ^m_{bu}	12.88 d^{-1}	12.88 d^{-1}	12.88 d^{-1}
$\gamma^m_{\text{H}_2\text{O}}$	0.01 d^{-1}	0.01 d^{-1}	0.01 d^{-1}

Table B.10: Transport coefficients. Definition of these parameters was based on information reported in the literature on percentage of water and SCFA absorption.

Parameters associated to microbial phenomena

Microbial adherence was not taken into account, thus a_i was set to zero. The shear coefficient for all microorganisms was set as $b_i = 0.08$ d^{-1}. The additional residence time τ_i was set to 1.0 d, to achieve high microbial density, as reported in the literature.

Parameters associated to component properties

Since the densities of SCFA and lactate are close to water density and since microorganisms are cells, mainly composed of water, the simplifying assumption that all components have the same density as water was adopted. So, $\rho_i = 1000$ g/l.

Parameter	Value	Parameter	Value
w_{bu}	88.10 g/mol	w_{la}	90.08 g/mol
w_{ac}	60.05 g/mol	$w_{\text{H}_2\text{O}}$	18.00 g/mol
w_{pro}	74.08 g/mol	w_{x_i}	113.00 g/mol

Table B.11: Molecular weights.

B.4. Reaction network, kinetics and stoichiometry

This section collects the biochemical material of Section 2.6: the reactions, the kinetic laws and the derivation of the yield factors from the fractions f and η.

Stoichiometric reactions

Table B.12 lists the twelve reactions behind the arrows of Figure 2.2. Each substrate has one or more catabolic reactions, which produce metabolites, and one anabolic reaction, which produces biomass.

Table B.12: Stoichiometric reactions used in the model. The biomass is assumed to have the empirical formula $C_5H_7O_2N$ [4].

Glucose utilization	
$C_6H_{12}O_6 \longrightarrow 2\,CH_3CHOHCOOH$	(R1)
$C_6H_{12}O_6 + 2\,H_2O \longrightarrow 2\,CH_3COOH + 2\,CO_2 + 4\,H_2$	(R2)
$3\,C_6H_{12}O_6 \longrightarrow 4\,CH_3CH_2COOH + 2\,CH_3COOH + 2\,CO_2 + 2\,H_2O$	(R3)
$C_6H_{12}O_6 \longrightarrow CH_3CH_2CH_2COOH + 2\,CO_2 + 2\,H_2$	(R4)
$5\,C_6H_{12}O_6 + 6\,NH_3 \longrightarrow 6\,C_5H_7O_2N + 18\,H_2O$	(R5)
Lactate utilization	
$3\,CH_3CHOHCOOH \longrightarrow 2\,CH_3CH_2COOH + CH_3COOH + CO_2 + H_2O$	(R6)
$2\,CH_3CHOHCOOH \longrightarrow CH_3CH_2CH_2COOH + 2\,H_2 + 2\,CO_2$	(R7)
$5\,CH_3CHOHCOOH + 3\,NH_3 \longrightarrow 3\,C_5H_7O_2N + 9\,H_2O$	(R8)
Hydrogen utilization: homoacetogenesis	
$4\,H_2 + 2\,CO_2 \longrightarrow CH_3COOH + 2\,H_2O$	(R9)
$10\,H_2 + 5\,CO_2 + NH_3 \longrightarrow C_5H_7O_2N + 8\,H_2O$	(R10)
Hydrogen utilization: methanogenesis	
$4\,H_2 + CO_2 \longrightarrow CH_4 + 2\,H_2O$	(R11)
$10\,H_2 + 5\,CO_2 + NH_3 \longrightarrow C_5H_7O_2N + 8\,H_2O$	(R12)

Processes and kinetic expressions

In the mass balances the reactions of Table B.12 are condensed into nine processes $j = 1, \ldots, 9$:

1. hydrolysis,

2. glucose utilization,
3. lactate utilization,
4. hydrogen utilization: homoacetogenesis,
5. hydrogen utilization: methanogenesis,
6. decay of x_{su},
7. decay of x_{la},
8. decay of $x_{\mathrm{H_2a}}$,
9. decay of $x_{\mathrm{H_2m}}$.

Hydrolysis follows the Contois equation, so its rate depends on the ratio between polysaccharides and the bacteria that hydrolyse them. Substrate uptake follows Monod kinetics. Methanogenesis is multiplied by an empirical inhibition function I_{pH}, which is the only pH effect kept in the model. Decay is first order, with the same constant k_d for the four groups.

Each of the first five rates is proportional to the concentration of the biomass that catalyses the process. We write this biomass as a generic argument χ, so that the same expressions can be reused in Chapter 3, where the biomass entering each process is redefined. The expressions are those of [31]:

$$\rho_1(\chi) = k_{\mathrm{hyd},z}\,\frac{z\,\chi}{K_{x,z}\chi + z}, \tag{B.1}$$

$$\rho_2(\chi) = k_{m,\mathrm{su}}\,\frac{s_{\mathrm{su}}\chi}{K_{s,\mathrm{su}} + s_{\mathrm{su}}}, \tag{B.2}$$

$$\rho_3(\chi) = k_{m,\mathrm{la}}\,\frac{s_{\mathrm{la}}\chi}{K_{s,\mathrm{la}} + s_{\mathrm{la}}}, \tag{B.3}$$

$$\rho_4(\chi) = k_{m,\mathrm{H_2a}}\,\frac{s_{\mathrm{H_2a}}\chi}{K_{s,\mathrm{H_2a}} + s_{\mathrm{H_2a}}}, \tag{B.4}$$

$$\rho_5(\chi) = k_{m,\mathrm{H_2m}}\,\frac{s_{\mathrm{H_2m}}\chi}{K_{s,\mathrm{H_2m}} + s_{\mathrm{H_2m}}}\,I_{\mathrm{pH}}, \tag{B.5}$$

$$\rho_6(\xi) = k_d x_{\mathrm{su}}, \tag{B.6}$$

$$\rho_7(\xi) = k_d x_{\mathrm{la}}, \tag{B.7}$$

$$\rho_8(\xi) = k_d x_{\mathrm{H_2a}}, \tag{B.8}$$

$$\rho_9(\xi) = k_d x_{\mathrm{H_2m}}, \tag{B.9}$$

with

$$I_{\mathrm{pH}} = \begin{cases} \exp\left(-3\left(\dfrac{pH - pH_{\mathrm{U}}}{pH_{\mathrm{U}} - pH_{\mathrm{L}}}\right)^2\right), & \text{if } pH < pH_{\mathrm{U}}, \\ 1, & \text{if } pH \geq pH_{\mathrm{U}}. \end{cases} \tag{B.10}$$

In the reference formulation each process is catalysed by a single group, so

$$\rho_1(x_{\mathrm{su}}), \quad \rho_2(x_{\mathrm{su}}), \quad \rho_3(x_{\mathrm{la}}), \quad \rho_4(x_{\mathrm{H_2a}}), \quad \rho_5(x_{\mathrm{H_2m}}), \tag{B.11}$$

and the notation $\rho_j(\xi)$ used in the mass balances stands for these values. In ρ_4 and ρ_5 the dissolved hydrogen is written $s_{\mathrm{H_2a}}$ and $s_{\mathrm{H_2m}}$, but both symbols denote the same variable $s_{\mathrm{H_2}}$: the extra letter only recalls which group consumes it.

Yield factors as functions of f and η

$$Y_{i,j} = f_j \sum_k \eta_{j,k}\, c_{i,k}, \qquad Y_j = c_x\,(1 - f_j)\,, \tag{B.12}$$

The yields are obtained from Table B.12 by applying the general structure (B.12), that is by multiplying the catabolic fraction f by the coefficients of each catabolic route, weighted by the η_k, and the anabolic coefficient by $1 - f$. Mass and elemental balances then hold by construction.

Glucose. Glucose feeds the catabolic reactions R1–R4 and the anabolic reaction R5. With f_{su} the catabolic fraction, R5 gives the biomass yield $Y_{\mathrm{su}} = \frac{6}{5}(1 - f_{\mathrm{su}})$, while the catabolic flux is split by η_k, $k = 1, \ldots, 4$, with $\eta_4 = 1 - (\eta_1 + \eta_2 + \eta_3)$. All the yields then read

$$Y_{\mathrm{la,su}} = 2 f_{\mathrm{su}} \eta_1, \qquad Y_{\mathrm{H_2,su}} = 4 f_{\mathrm{su}} \eta_2 + 2 f_{\mathrm{su}} \eta_4, \tag{B.13}$$

$$Y_{\mathrm{ac,su}} = 2 f_{\mathrm{su}} \eta_2 + \tfrac{2}{3} f_{\mathrm{su}} \eta_3, \qquad Y_{\mathrm{CO_2,su}} = 2 f_{\mathrm{su}} \eta_2 + \tfrac{2}{3} f_{\mathrm{su}} \eta_3 + 2 f_{\mathrm{su}} \eta_4, \tag{B.14}$$

$$Y_{\mathrm{pro,su}} = \tfrac{4}{3} f_{\mathrm{su}} \eta_3, \qquad Y_{\mathrm{bu,su}} = f_{\mathrm{su}} \eta_4, \tag{B.15}$$

$$Y_{\mathrm{H_2O,su}} = -2 f_{\mathrm{su}} \eta_2 + \tfrac{2}{3} f_{\mathrm{su}} \eta_3 + \tfrac{18}{5}\,(1 - f_{\mathrm{su}})\,, \tag{B.16}$$

so that 8 parameters are reduced to 4.

Lactate. Lactate follows two catabolic routes: R6, which produces propionate and acetate, and R7, which produces butyrate. With f_{la} the catabolic fraction and $\eta_{\mathrm{la},1}$ the

share going to R6, so that $\eta_{\mathrm{la},2} = 1 - \eta_{\mathrm{la},1}$, the anabolic reaction R8 gives $Y_{\mathrm{la}} = \frac{3}{5}(1 - f_{\mathrm{la}})$ and the other yields are

$$Y_{\mathrm{ac,la}} = \tfrac{1}{3} f_{\mathrm{la}}\, \eta_{\mathrm{la},1}, \qquad Y_{\mathrm{bu,la}} = \tfrac{1}{2} f_{\mathrm{la}}\, \eta_{\mathrm{la},2}, \tag{B.17}$$

$$Y_{\mathrm{pro,la}} = \tfrac{2}{3} f_{\mathrm{la}}\, \eta_{\mathrm{la},1}, \qquad Y_{\mathrm{H_2,la}} = f_{\mathrm{la}}\, \eta_{\mathrm{la},2}, \tag{B.18}$$

$$Y_{\mathrm{CO_2,la}} = \tfrac{1}{3} f_{\mathrm{la}}\, \eta_{\mathrm{la},1} + f_{\mathrm{la}}\, \eta_{\mathrm{la},2}, \qquad Y_{\mathrm{H_2O,la}} = \tfrac{1}{3} f_{\mathrm{la}}\, \eta_{\mathrm{la},1} + \tfrac{9}{5}\,(1 - f_{\mathrm{la}})\,, \tag{B.19}$$

reducing 6 parameters to 2.

Hydrogen. Two groups compete for the same dissolved pool $s_{\mathrm{H_2}}$: homoacetogenic bacteria through R9–R10 and methanogenic archaea through R11–R12. Each group has a single catabolic route, so one parameter per group is enough, and the two anabolic reactions are identical. With $f_{\mathrm{H_2a}}$ and $f_{\mathrm{H_2m}}$ the catabolic fractions,

$$\begin{aligned}
Y_{\mathrm{H_2a}} &= \tfrac{1}{10}\,(1 - f_{\mathrm{H_2a}})\,, & Y_{\mathrm{H_2m}} &= \tfrac{1}{10}\,(1 - f_{\mathrm{H_2m}})\,,\\
Y_{\mathrm{ac,H_2a}} &= \tfrac{1}{4}\, f_{\mathrm{H_2a}}, & Y_{\mathrm{CH_4,H_2m}} &= \tfrac{1}{4}\, f_{\mathrm{H_2m}},\\
Y_{\mathrm{CO_2,H_2a}} &= -\tfrac{1}{2}\, f_{\mathrm{H_2a}} - \tfrac{1}{2}\,(1 - f_{\mathrm{H_2a}})\,, & Y_{\mathrm{CO_2,H_2m}} &= -\tfrac{1}{4}\, f_{\mathrm{H_2m}} - \tfrac{1}{2}\,(1 - f_{\mathrm{H_2m}})\,,\\
Y_{\mathrm{H_2O,H_2a}} &= \tfrac{1}{2}\, f_{\mathrm{H_2a}} + \tfrac{4}{5}\,(1 - f_{\mathrm{H_2a}})\,, & Y_{\mathrm{H_2O,H_2m}} &= \tfrac{1}{2}\, f_{\mathrm{H_2m}} + \tfrac{4}{5}\,(1 - f_{\mathrm{H_2m}})\,.
\end{aligned} \tag{B.20}$$

In the balance of dissolved hydrogen both ρ_4 and ρ_5 enter with coefficient -1: one mole of reaction consumes one mole of $s_{\mathrm{H_2}}$, whatever the later split between catabolism and anabolism. The balances therefore hold for any $f_{\mathrm{H_2a}}, f_{\mathrm{H_2m}} \in [0, 1]$, with no cross-constraint between the two groups.

Petersen matrices

The element (j, i) of the matrix is the yield factor of component i in process j. Empty entries are zero.

		Component $\rightarrow i$									
j	**Process** $\downarrow$	1	2	3	4	5	6	7	8	9	
		s_{su}	s_{la}	$s_{\mathrm{H_2}}$	s_{ac}	s_{pro}	s_{bu}	$s_{\mathrm{CH_4}}$	$s_{\mathrm{CO_2}}$	$s_{\mathrm{H_2O}}$	**Kinetic rate**
1	Hydrolysis	$Y_{\mathrm{su},z}$									$\rho_1(\xi)$
2	Glucose utilization	-1	$Y_{\mathrm{la,su}}$	$Y_{\mathrm{H_2,su}}$	$Y_{\mathrm{ac,su}}$	$Y_{\mathrm{pro,su}}$	$Y_{\mathrm{bu,su}}$		$Y_{\mathrm{CO_2,su}}$	$Y_{\mathrm{H_2O,su}}$	$\rho_2(\xi)$
3	Lactate utilization		-1	$Y_{\mathrm{H_2,la}}$	$Y_{\mathrm{ac,la}}$	$Y_{\mathrm{pro,la}}$	$Y_{\mathrm{bu,la}}$		$Y_{\mathrm{CO_2,la}}$	$Y_{\mathrm{H_2O,la}}$	$\rho_3(\xi)$
4	Hydrogen utilization: homoacetogenesis			-1	$Y_{\mathrm{ac,H_2a}}$				$Y_{\mathrm{CO_2,H_2a}}$	$Y_{\mathrm{H_2O,H_2a}}$	$\rho_4(\xi)$
5	Hydrogen utilization: methanogenesis			-1				$Y_{\mathrm{CH_4,H_2m}}$	$Y_{\mathrm{CO_2,H_2m}}$	$Y_{\mathrm{H_2O,H_2m}}$	$\rho_5(\xi)$

Table B.13: Petersen matrix of the biochemical reactions: soluble components.

j	**Process** $\downarrow$	**Component** $\rightarrow i$					
		10	11	12	13	14	
		z	x_{su}	x_{la}	$x_{\mathrm{H_2a}}$	$x_{\mathrm{H_2m}}$	**Kinetic rate**
1	Hydrolysis	-1					$\rho_1(\xi)$
2	Glucose utilization		Y_{su}				$\rho_2(\xi)$
3	Lactate utilization			Y_{la}			$\rho_3(\xi)$
4	Hydrogen utilization: homoacetogenesis				$Y_{\mathrm{H_2a}}$		$\rho_4(\xi)$
5	Hydrogen utilization: methanogenesis					$Y_{\mathrm{H_2m}}$	$\rho_5(\xi)$
6	Decay of x_{su}		-1				$\rho_6(\xi)$
7	Decay of x_{la}			-1			$\rho_7(\xi)$
8	Decay of $x_{\mathrm{H_2a}}$				-1		$\rho_8(\xi)$
9	Decay of $x_{\mathrm{H_2m}}$					-1	$\rho_9(\xi)$

Table B.14: Petersen matrix of the biochemical reactions: particulate components. In the lumen, z is the aggregated concentration of polysaccharides from dietary origin plus the mucins coming from the mucus. In the mucus, z is the concentration of mucins originating from the host.

B.5. Simulation settings

The input flow rate q_{in} was set to 1.5 l/d. Exogenous polysaccharides are considered as fibers, and fiber intake was modelled as a periodical pulse, with three meals of 30 min per day; the amplitude of the pulse is calculated according to the desired fiber intake. Initial conditions were determined by simulating the model with a total periodic feed of 10 g of fiber per day until stationary state was reached in the proximal and transverse colon. Mucin production was set as $\Gamma = 5$ g/d, equally split among the three sections.

The volume in the liquid phase of the distal colon was initialized at $V_l = 0.100$ l. The volume of one excretion was set at 0.200 l, so that when the volume occupied by the chyme in the lumen of the distal colon reaches 0.300 l it is reset to 0.100 l. The dilution rate of the small-intestine cascade in (2.19) is $D = 9\ \mathrm{d}^{-1}$.

C | Numerical implementation

This appendix details the Python implementation of the model of Chapter 2, its verification against the original `Matlab/Simulink` version of [31], and the performance obtained through just-in-time compilation.

C.1. Structure of the code

The implementation is organized in four blocks:

- a function returning the right-hand side $f(\mathbf{y}, \theta, t)$ of the 106 equations;
- a parameter structure holding θ, in which the yield factors are recomputed from θ_{stoich} (Section 2.5) at every parameter change;
- a solver interface handling burn-in, scenario runs and excretion events;
- post-processing routines for the extraction of the outputs of interest.

The state vector stacks, for each of the six colonic compartments, the 17 states of (2.1), followed by the three states of the small-intestine cascade (2.19) and by the luminal volume V_l of the distal colon.

C.2. Integration

- **Solver**: `scipy.integrate.solve_ivp`, method `BDF`, chosen for the stiffness induced by the separation between reaction and transport time scales; it plays the role of `ode15s` in the original implementation.
- **Excretion events**: the condition $V_l = V_{\max}$ is detected through the event mechanism of the solver; the integration is stopped, V_l is reset to V_{reset} at unchanged concentrations, and restarted.
- **Simulation protocol**: a *burn-in* phase brings the system to its periodic regime, a *scenario* phase produces the trajectories actually used.

C.3. Libraries

- **NumPy**: array algebra and vectorized evaluation of the reaction terms;
- **SciPy**: time integration and event handling;
- **Numba**: just-in-time compilation of the right-hand side;
- **Matplotlib**: visualization of the results.

C.4. Verification

The Python implementation has been verified by reproducing the simulations published in [31] under identical conditions: same parameter values, same initial conditions and same inputs (dietary polysaccharide profile and flow rates). The figures of the original paper have been replicated one by one, covering substrate dynamics, microbial populations, metabolite production and gas evolution. The curves obtained coincide with the published ones; the residual differences are not visible on the scale of the figures and are attributable to the different solver implementations, to the tolerance settings and to the treatment of the excretion events. The figures are therefore not reproduced here.

C.5. Acceleration with Numba

The cost of a single simulation is dominated by the repeated evaluation of the right-hand side: 106 states, nonlinear kinetics and a large number of internal steps imposed by stiffness. The function is therefore compiled with the `@numba.njit` decorator, which required avoiding Python objects inside the right-hand side, using typed NumPy arrays of fixed size, and removing dynamic memory allocation from the integration loop.

Table C.1 compares the pure Python implementation with the compiled one on the same simulation.

Metric	No JIT	With JIT
Warm-up time (s)	0.000	0.602
Burn-in phase (s)	26.379	3.936
Scenario phase (s)	14.393	1.810
Total simulation time (s)	40.772	5.746
Total wall time (s)	40.773	6.348

Table C.1: Computational performance with and without `Numba`.

The total simulation time drops from 40.8 s to 5.7 s, a speedup of about 7.1×. The compilation overhead of the warm-up phase is paid only once per session and becomes irrelevant as soon as several simulations are performed, which is precisely the regime of the sensitivity analysis, of the uncertainty quantification and of the inverse problem, where thousands of model evaluations are required.

D | Sensitivity

This appendix reports the complete set of factors used in the sensitivity analysis and a few implementation details. The ranges are built as described in Section 4.2, and the compositional mapping is derived in Appendix A.

D.1. Input factors

Table D.1 lists the 30 factors used in the Morris screening. The factors are labelled directly by the model parameters they perturb. The compositional parameters are handled in two ways. For the multi-part compositions $\boldsymbol{\eta}$ and $\boldsymbol{\alpha}$, the labels η_i and α_i identify the independent log-perturbation associated with each component; the physical components themselves are obtained after closure and are therefore not independent, and the range reported in the table is that of the underlying log-factor. For the two-part proportions $\eta_{la,1}$ and β_1 a single log-factor is enough, so the reported range is that of the proportion itself. The nominal values η_i°, α_i° and β_i° are the values used in the model.

Table D.1: Input factors of the Morris screening. For ordinary positive quantities, $\times/\div$ 2 denotes log-uniform sampling in $[\theta_j^\circ/2,\, 2\theta_j^\circ]$. For two-part proportions, the reported interval is the one induced on the proportion by a uniform log-factor in $[-\log 2,\, \log 2]$. For multi-part compositions, the factor name identifies the component being perturbed and the reported range is that of its underlying log-factor. Units follow Muñoz-Tamayo et al. [31].

Factor	Nominal	Range	Rule
Catabolic fractions			
f_{su}	0.90	0.818–0.947	log-factor $\pm \log 2$
f_{la}	0.80	0.667–0.889	log-factor $\pm \log 2$
f_{h2a}	0.57	0.399–0.726	log-factor $\pm \log 2$
f_{h2m}	0.38	0.235–0.551	log-factor $\pm \log 2$
Maximum specific rates			

continued on next page

Table D.1 – continued from previous page

Factor	Nominal	Range	Rule
$k_{m,su}$	7.92	3.96–15.84	$\times/\div 2$
$k_{m,la}$	103.0	51.5–206.0	$\times/\div 2$
$k_{m,h2a}$	108.84	54.42–217.68	$\times/\div 2$
$k_{m,h2m}$	22.58	11.29–45.16	$\times/\div 2$
$k_{m,rs} = k_{m,mu}$	1199.69	599.85–2399.38	$\times/\div 2$
Retention and mucus interaction			
τ (all four groups)	1.0 d	0.5–2.0 d	$\times/\div 2$
a (adhesion)	0.01 d^{-1}	0.005–0.02 d^{-1}	$\times/\div 2$
b (shear)	0.08 d^{-1}	0.04–0.16 d^{-1}	$\times/\div 2$
Decay rates			
k_{dec,X_1}	0.01 d^{-1}	0.001–0.1 d^{-1}	± 1 decade
k_{dec,X_2}	0.01 d^{-1}	0.001–0.1 d^{-1}	± 1 decade
k_{dec,X_3}	0.01 d^{-1}	0.001–0.1 d^{-1}	± 1 decade
k_{dec,X_4}	0.01 d^{-1}	0.001–0.1 d^{-1}	± 1 decade
SCFA absorption			
$\gamma^{P}_{\mathrm{SCFA}}$	2.526	1.263–5.052	$\times/\div 2$
$\gamma^{T}_{\mathrm{SCFA}}$	1.200	0.600–2.400	$\times/\div 2$
$\gamma^{D}_{\mathrm{SCFA}}$	2.418	1.209–4.836	$\times/\div 2$
Inputs			
Fibre intake	25 g/d	12.5–50 g/d	$\times/\div 2$
Mucin secretion	5 g/d	2.5–10 g/d	$\times/\div 2$
Compositional factors, multi-part			
η_1 (lactate)	0.25	$[-\log 2, \log 2]$	log-factor + closure
η_2 (acetate)	0.25	$[-\log 2, \log 2]$	log-factor + closure
η_3 (propionate)	0.20	$[-\log 2, \log 2]$	log-factor + closure
η_4 (butyrate)	0.30	$[-\log 2, \log 2]$	log-factor + closure
α_1 (glucose uptake)	0.50	$[-\log 2, \log 2]$	log-factor + closure
α_2 (lactate uptake)	0.30	$[-\log 2, \log 2]$	log-factor + closure
α_3 (homoacetogenesis)	0.20	$[-\log 2, \log 2]$	log-factor + closure
Compositional factors, two-part			
$\eta_{la,1}$ (R6 vs. R7 split)	0.50	0.333–0.667	log-factor $\pm \log 2$
β_1 (lactate uptake)	0.80	0.667–0.889	log-factor $\pm \log 2$

D.2. Implementation

The sampling and the computation of the indices are implemented in Python with SALib [17]. The parameters generated by the design are first transformed to their physical values. In particular, the factors labelled η_i and α_i are mapped to valid compositions through Eq. (A.8). The model evaluations are independent and are run in parallel, and their outputs are collected in the original sampling order before the indices are computed.

For the Morris screening we use a grid with $L = 4$ levels and select 60 trajectories from 200 candidates with the optimized design of Campolongo et al. [10]: with 30 factors this gives $60(30 + 1) = 1860$ evaluations. The Sobol' analysis is then applied to the twelve retained factors only, keeping the same input ranges, with the first-, second- and total-order indices estimated by the scheme of Saltelli et al. [43] with $N = 2048$, that is $N(2k + 2) = 53\,248$ evaluations for $k = 12$.

D.3. Second-order Sobol' indices

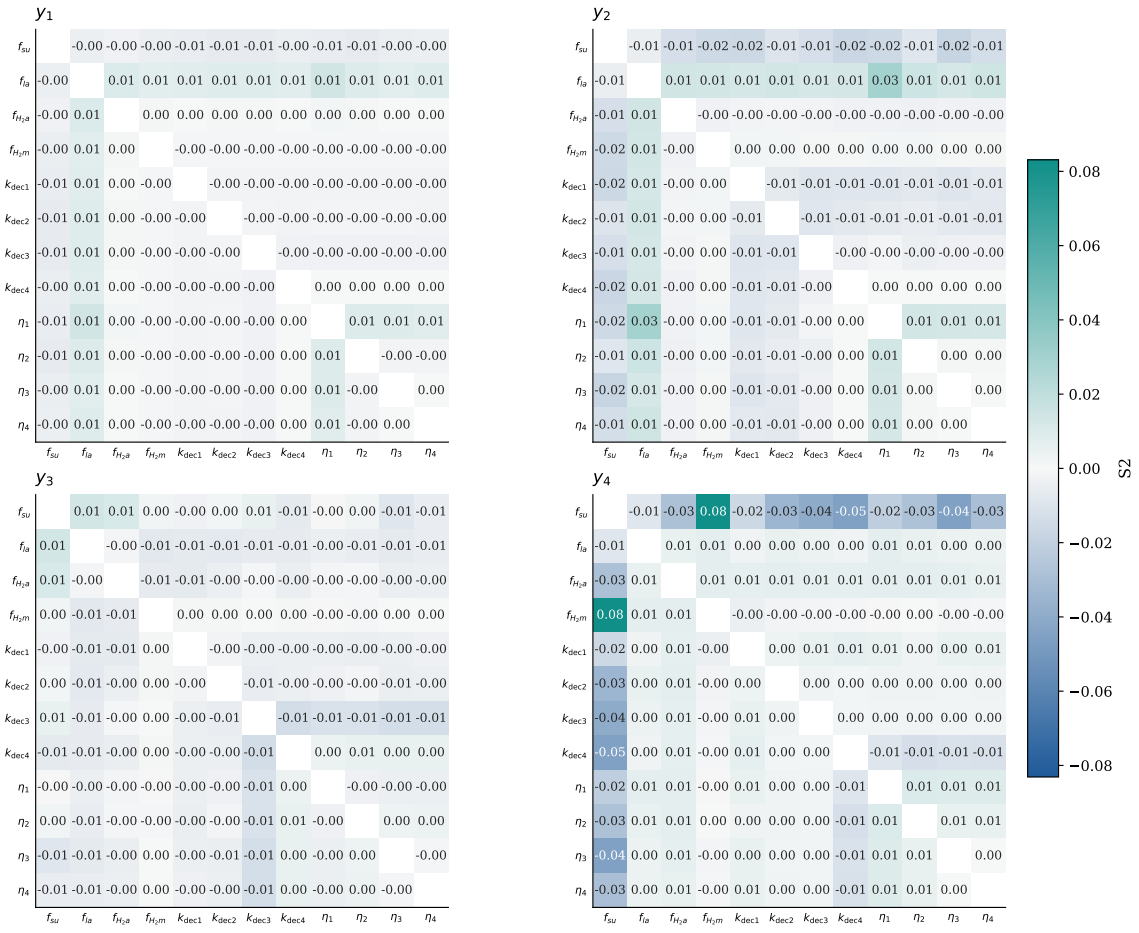


Figure D.1: Second-order Sobol' indices S_2 of the twelve retained factors on the four distal fractions, from the same design as Figure 4.2. The largest entry is the pair (f_{su}, f_{h2m}) on y_4; the remaining entries are of the same order as the noise of the estimator, whose negative values reach -0.04.

E | Inverse problem

This appendix collects the design of the training set of the surrogate, its training settings and accuracy plot, the implementation and the settings of the sampler, the comparison between samplers, the check of the inversion against the full model, and the implementation of the cohort run.

E.1. Low-discrepancy design

The training points of Section 5.3 come from a quasi-Monte Carlo design. Quasi-Monte Carlo replaces the independent random draws of Monte Carlo by a deterministic low-discrepancy sequence [32, 45]. The star discrepancy of a point set $P = \{x_1, \dots, x_N\} \subset [0,1]^d$ is

$$D_N^*(P) = \sup_B \left| \frac{\#\{x_i \in B\}}{N} - \mathrm{vol}(B) \right|,$$

the supremum being taken over all axis-aligned boxes B anchored at the origin. It measures how far the point set is from filling the domain uniformly: for every such box it compares the fraction of points inside with the volume of the box, and takes the worst case. By the Koksma–Hlawka inequality the integration error of a function of bounded variation $V(g)$ is bounded by $V(g)\, D_N^*(P)$, so a smaller discrepancy means a more even coverage. Random sampling gives $D_N^* \sim N^{-1/2}$, a low-discrepancy sequence gives $D_N^* \sim (\log N)^d / N$ [8]. For the same number of expensive model runs the design therefore leaves fewer gaps in the parameter box. The sequence is also nested: a future run with 2^{16} points contains all the points used here, so the design can be extended without repeating any simulation.

E.2. Training of the surrogate

Architecture. The inputs are rescaled by a min–max map to $[-1, 1]$ and passed to a multilayer perceptron with four hidden layers of 128 units and tanh activations (50 692 parameters). The final linear layer has four outputs, which are mapped to the simplex by a softmax (Figure E.1). The network is trained using the squared Aitchison distance

between the predicted and simulated compositions,

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \left\| \operatorname{clr}(\hat{\mathbf{y}}(f_i)) - \operatorname{clr}(\mathbf{y}_i) \right\|_2^2 .$$

The architecture is deliberately plain, since the map from the four parameters to the output composition is smooth.

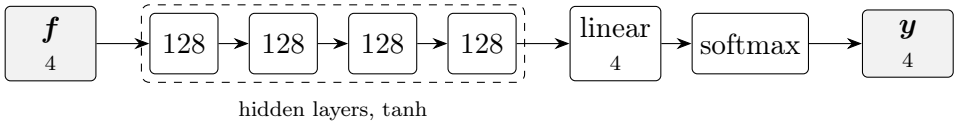


Figure E.1: Structure of the surrogate. The four catabolic fractions enter the network after a min–max rescaling and the softmax returns a composition.

Training. The design points are split at random, 80 % for training, 10 % for validation and 10 % for test. The network is trained for 2000 epochs with Adam [22], batch size 1024, and a cosine learning-rate schedule [25] decreasing from $1.5 \cdot 10^{-3}$ to 10^{-5}; the weights with the lowest validation loss are kept, which here happens at epoch 1485. The whole training takes about nine minutes, negligible next to the cost of generating the design.

Implementation. The network is implemented and trained in PyTorch, with NumPy for the data preparation and the metrics. A fixed seed is used to make the training reproducible. The trained network is also exported to TorchScript.

	y_1	y_2	y_3	y_4
R^2 (clr)	0.99980	0.99861	0.99986	0.99914
log-ratio RMSE	0.0079	0.0808	0.0325	0.0611

mean	median	p95
0.0254	0.0121	0.0601

Table E.1: Surrogate accuracy on the test set. Top: coefficient of determination in clr coordinates and root mean squared error of the log-ratios, per microbial group. Bottom: Aitchison distance between surrogate and simulation.

E.3. Check against the full model

The inversion uses the surrogate everywhere, so the fit it produces is a statement about the surrogate until the full model is put back. For one hundred subjects drawn at random

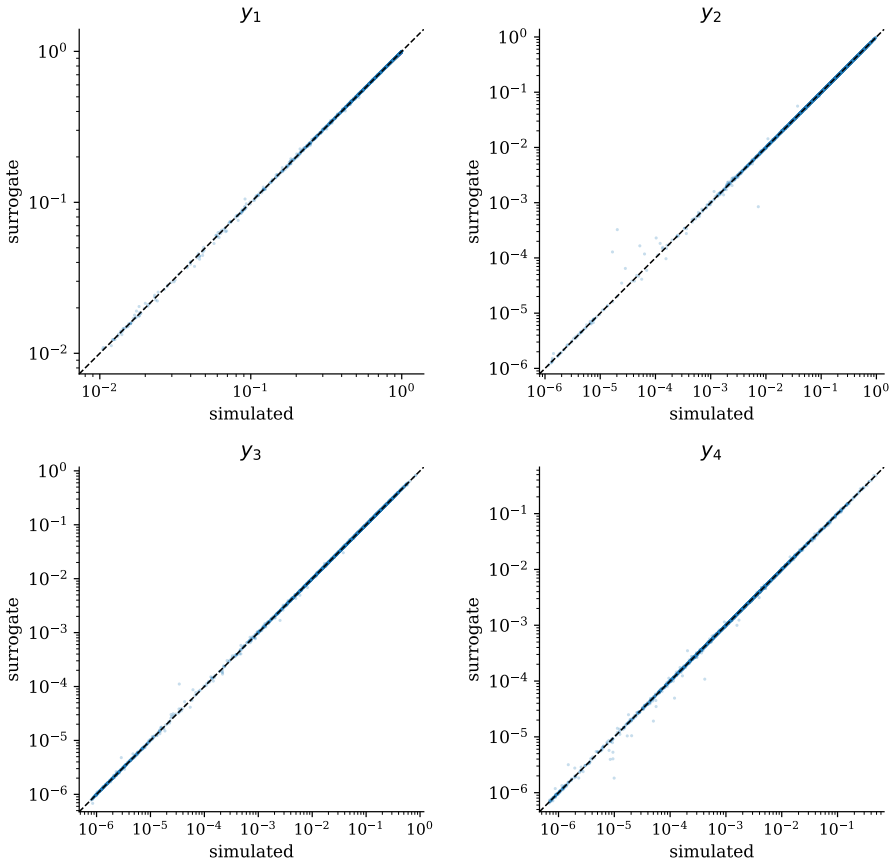


Figure E.2: Surrogate predictions against the simulated fractions on the test set. Each panel is one microbial group, on logarithmic axes, with the identity line dashed.

from the cohort we take the posterior median of the four catabolic fractions, integrate the full model to its steady state at those values with the same convergence protocol used to build the training set, and compute the Aitchison distance between the simulated distal composition and the observed one. This is the only point of the chapter where the surrogate is closed back onto the model it approximates, and the reason for restricting it to a subsample is cost: each re-simulation is a full integration, whereas the inversion itself never runs one. The resulting distances are reported in Table 5.1.

E.4. Implementation of the sampler

The posterior is built in PyMC [1]. The four catabolic fractions are declared as truncated Gaussian variables on $[0.01, 0.99]$ as in Eq. (5.8), and the log-likelihood (5.7) is added as a potential term. The surrogate is loaded from its exported TorchScript file and evaluated through a custom operator, so a proposal costs one forward pass of the network and the mechanistic model is never integrated inside the chain.

The step is DEMetropolisZ with its default differential-evolution weight λ, an additive jitter of scale 10^{-3}, and λ tuned every 100 iterations during the tuning phase, the archive of past states dropping its oldest 90 % at the end of tuning. Tuning draws are discarded. Each run uses a fixed seed, and the chains start from prescribed points spread over the box rather than from random draws, so that a failure to mix is visible as a disagreement between chains. Traces are stored in NetCDF format; $\hat{R}$, effective sample sizes and highest-density intervals are computed with ArviZ.

	representative sample	cohort
Cases inverted	1	3613
Chains	6	4
Tuning steps	$5 \cdot 10^4$	8000
Draws per chain	10^5	8000
Starting points	6 spread over the box	4 spread over the box

Table E.2: Sampling settings of the two runs of Chapter 5.

E.5. Sampler comparison

For the medoid sample, DEMetropolisZ was compared with NUTS using the same surrogate, likelihood and priors. NUTS produces less correlated samples, but each draw requires the integration of a trajectory and is computationally more expensive.

	DEMetropolisZ	NUTS
Mean $\hat{R}$	1.00	1.01
ESS / draw	0.012	0.051
ESS / second	51.6	1.55
Divergences	–	1

Table E.3: Comparison between DEMetropolisZ and NUTS for the medoid sample.

NUTS is more efficient per draw, while DEMetropolisZ gives about thirty times more effective samples per unit time. For this reason DEMetropolisZ is used for all the inversions of this work.

E.6. Implementation of the cohort run

The 3613 cleaned compositions of Section 1.2 are inverted here; the two samples whose first profile was clipped at the detection floor were already removed there, a clipped w_1 not being a quantitative target for the clr likelihood. The inversions are independent and are run in parallel, sixteen at a time on 64 cores, each case keeping its four chains on four cores and using its own seed so that the runs are reproducible. The settings of each case are those of Table E.2. The full cohort takes 2 h 46 min of wall-clock time, that is about 2.8 s per sample of amortised cost and roughly 45 s per individual inversion. Convergence diagnostics ($\hat{R}$, effective sample size, divergences), posterior summaries and predictive checks are stored for every case and are the source of the numbers reported in Section 5.5. All 3613 cases ran to completion, none failed.

List of Figures

1 The two levels of description used by the model. (a) The ascending, transverse and descending parts, with the sigmoid colon and the rectum, are grouped into three regions in series. (b) Within each region, the lumen carries the flowing content while the wall is covered by a mucus layer that hosts attached bacteria. Adapted from [27]. 3

1.1 Composition of the datasets of Labarthe et al. [23] by clinical group. *Train* is not a cohort but the pooled training set on which the four profiles were inferred (7 studies, 5 countries, balanced in health status); the other five are the external validation cohorts, each selected for one specific effect: *CD cohort*, Crohn's disease; *HMP2*, second Human Microbiome Project (IBD); *MetaCardis*, cardiometabolic cohort; *Med. diet*, Mediterranean-diet intervention; *Parkinson*, Parkinson's disease. The control arm of the Mediterranean-diet trial is counted in *Control*. 6

1.2 Frequency of the 101 AFTs inside each functional profile [23]. Profile 1 is the most hydrolytic, with the largest share of glycoside hydrolases; profiles 2, 3 and 4 are shifted towards the fermentation of simple sugars, and profile 4 is the only one carrying methanogenesis. 7

1.3 Distribution of the four profiles over the pooled cohort, with the retained region shaded. 8

2.1 Structure of the model. Three regions in series, each split into a lumen and a mucus compartment; the lumen carries the flow from the upper digestive tract to excretion. The transverse region is drawn in detail: in both phases the state variables z, $\mathbf{x}$, $\mathbf{s}$, $\mathbf{s}_g$ are linked by the reactions and by the liquid–gas transfer Q. Lumen and mucus exchange soluble components (γ^l), biomass (adherence a, detachment b) and mucins (q_o^m); the host secretes mucins (Γ) and absorbs metabolites (γ^m). 12

2.2 Aggregated metabolic pathway, the graphical counterpart of the reactions of Table B.12 (Appendix B.4). Each arrow is labelled by the microbial group that catalyses it: x_{su} hydrolyses polysaccharides into glucose and ferments glucose into lactate, short chain fatty acids and gases; x_{la} turns lactate into short chain fatty acids and gases; $x_{\mathrm{H_2a}}$ and $x_{\mathrm{H_2m}}$ consume hydrogen and produce acetate and methane respectively. Once dissolved, H_2, CH_4 and CO_2 also move between the liquid and the gas phase of the compartment. 17

2.3 The single arrow "lactate utilization" of Figure 2.2, split into its catabolic and anabolic parts. The fraction f_{la} of the lactate taken up is degraded, the fraction $1 - f_{\mathrm{la}}$ becomes biomass, and $\eta_{\mathrm{la},1}$ splits the catabolic flux between the two degradation routes. Gases and water are omitted; R6–R8 are the reactions of Table B.12. 19

3.1 Top: the four distal biomasses $x^{(j)}$ recorded at each excretion, in g/L. Bottom: the windowed total variation V_n of Eq. (3.10), in logarithmic scale, against the threshold $\varepsilon = 10^{-3}$ (horizontal dashed line). Since V_n sums the changes over a window of length $W = 30$ d, it is defined only from day 30 on; the green dotted line marks the excretion at which it falls below ε. 27

3.2 Distal biomass of the four groups, in g/L, for $n = 1000$ initial conditions. Each of the 106 state variables is multiplied by an independent factor drawn uniformly in $[1-\ell,\ 1+\ell]$ with $\ell = 0.5$, that is a perturbation of $\pm 50\%$ around the baseline value; all states are perturbed, although only the biomasses are shown here. In each panel the left box is the distribution of the distal biomass imposed at $t = 0$, with the individual samples shown as points, and the right box is the distribution obtained at the last excretion. The right boxes are collapsed to a line at the scale of the figure. 28

4.1 Morris index μ^* of the 30 factors, for each of the four distal fractions. Grid with $L = 4$ levels and step $\Delta = 2/3$; 60 trajectories selected out of 200 candidates by maximizing their mutual spread [10], for 1860 simulations. . 34

4.2 First-order (S_1, light blue) and total (S_T, dark blue) Sobol' indices of the twelve retained factors on the four distal fractions, sorted by S_T. Sobol' sequence and the estimator of [43] extended to the second order, $N(2k+2)$ runs with $N = 2048$ and $k = 12$, i.e. 53 248 simulations; error bars are bootstrap confidence intervals. The second-order indices are reported in Figure D.1. 36

5.1 Posterior predictive distribution for the representative sample. The shaded region is the 95 % interval, the black line is the predictive median and the red line is the observed composition. 45

D.1 Second-order Sobol' indices S_2 of the twelve retained factors on the four distal fractions, from the same design as Figure 4.2. The largest entry is the pair (f_{su}, f_{h2m}) on y_4; the remaining entries are of the same order as the noise of the estimator, whose negative values reach -0.04. 80

E.1 Structure of the surrogate. The four catabolic fractions enter the network after a min–max rescaling and the softmax returns a composition. 82

E.2 Surrogate predictions against the simulated fractions on the test set. Each panel is one microbial group, on logarithmic axes, with the identity line dashed. 83

List of Tables

1.1 The cleaned dataset (3613 samples). All values are percentages: the first column gives the share of samples in which the profile is absent, the other two are computed on the remaining samples only. 9

5.1 Aitchison distances. The first row is the distance between a posterior predictive draw and the observation, over the complete cleaned dataset: it measures the width of the predictive distribution, whose scale is set by the tolerance σ of (5.7). The second row is the distance between model and data at the posterior median of the parameters, for one hundred randomly selected samples re-simulated with the full mechanistic model. The two rows are different quantities and should not be read as a comparison. . . . 46

B.1 State variables. 60
B.2 Parameters. Note that, as in [31], the symbol ρ is used both for the densities ρ_i and for the kinetic rates $\rho_j(\xi)$; the two are distinguished by the index, i running over components and j over processes. 61
B.3 Functions. 61
B.4 Subscripts and superscripts. 62
B.5 Physiological parameters. 62
B.6 Approximate dimensions of the human colon. [a]The volume of the mucus (v_m+v_g) is calculated assuming cylindric shape as $v_m+v_g = 0.25\pi\left(\phi^2 - (\phi - 2e_m)^2\right) L$. The thickness of the mucus e_m is taken to be equal approximately to 0.0830 cm; ϕ is the diameter of the colon (5 cm), and L is the length of each section. The volume of the lumen is calculated as $0.25\pi\phi^2 L - (v_m + v_g)$. 63
B.7 Physiological parameters. 64
B.8 Kinetic parameters and yield factors. Kinetic parameters are the same in all compartments. 65
B.9 Liquid–gas transfer parameters. 65

B.10 Transport coefficients. Definition of these parameters was based on information reported in the literature on percentage of water and SCFA absorption. 66
B.11 Molecular weights. 66
B.12 Stoichiometric reactions used in the model. The biomass is assumed to have the empirical formula $C_5H_7O_2N$ [4]. 67
B.13 Petersen matrix of the biochemical reactions: soluble components. 70
B.14 Petersen matrix of the biochemical reactions: particulate components. In the lumen, z is the aggregated concentration of polysaccharides from dietary origin plus the mucins coming from the mucus. In the mucus, z is the concentration of mucins originating from the host. 71

C.1 Computational performance with and without `Numba`. 74

D.1 Input factors of the Morris screening. For ordinary positive quantities, $\times/\div$ 2 denotes log-uniform sampling in $[\theta_j^\circ/2,\, 2\theta_j^\circ]$. For two-part proportions, the reported interval is the one induced on the proportion by a uniform log-factor in $[-\log 2,\, \log 2]$. For multi-part compositions, the factor name identifies the component being perturbed and the reported range is that of its underlying log-factor. Units follow Muñoz-Tamayo et al. [31]. 77

E.1 Surrogate accuracy on the test set. Top: coefficient of determination in clr coordinates and root mean squared error of the log-ratios, per microbial group. Bottom: Aitchison distance between surrogate and simulation. . . . 82
E.2 Sampling settings of the two runs of Chapter 5. 84
E.3 Comparison between DEMetropolisZ and NUTS for the medoid sample. . . 85